

Education and growth

The Solow model with human capital

The Solow model of Chapter 5 performed quite well by empirical standards, but we identified two systematic discrepancies between the international evidence and the predictions of the model: the steady state of the model seemed to substantially *underestimate* how strongly savings and population growth rates influence GDP per person, and the model seemed to substantially *overestimate* the speed by which economies converge to their steady states.

Is there a single modification of the Solow model that can redress *both* of its two empirical weaknesses? Indeed there is, and the idea behind it is brilliant. It is due to the economists, N. Gregory Mankiw, David Romer and David N. Weil, who presented it in an article entitled 'A Contribution to the Empirics of Economic Growth'.¹ We will now explain the idea intuitively.

Consider the problem concerning the speed of convergence. In the Solow model convergence is slower the larger the elasticity of output with respect to capital, α . We explained this fact in Chapter 5 where we found the rate of convergence to be $\lambda = (1 - \alpha)(n + g + \delta)$. If α could be chosen freely, we could simply set α sufficiently large to bring λ in accordance with the empirics, but α is one minus labour's share, so realistically α cannot be set much higher than $\frac{1}{3}$. However, suppose there were also another kind of capital with an output elasticity and 'income share' of its own, ϕ say, and that this capital were accumulated in the same way as physical capital (and therefore delayed convergence similarly). Furthermore, suppose the income share of this form of capital were accruing to workers. Then, for α maintained around $\frac{1}{3}$, one would have a slower convergence since $\alpha + \phi > \frac{1}{3}$, and still a labour share, $1 - \alpha$, around $\frac{2}{3}$.

This suggests the idea of adding *human capital* to the model: each year there is a certain stock of human capital, which can be thought of as the sum of all the education and training the workers have had, and this stock (per worker) serves to increase the productivity of workers. Just like physical capital, total human capital is assumed to accumulate by a certain exogenous fraction of GDP being added to it each year. The income share of human capital goes to workers, because human capital (embedded education) is linked to workers and its services therefore sold inseparably together with work hours.

Since human capital accumulates like physical capital, the inclusion of it will lower the model's speed of convergence. Because the output elasticity of physical capital is

1. N. Gregory Mankiw, David Romer and David N. Weil, 'A Contribution to the Empirics of Economic Growth', *The Quarterly Journal of Economics*, 107, 1992, pp. 407–437.

unchanged, the owners of physical capital will earn the same fraction of GDP. The fraction earned by human capital goes to the workers, so labour's share will also be unchanged. Thus, adding human capital should work to overcome the second of the empirical problems.

What about the first problem concerning the model's steady state prediction? Assume that the rate of investment in physical capital increases. This will create higher GDP per person in steady state because more physical capital per person is accumulated. If a certain fraction of GDP is invested in human capital, then the higher GDP first caused by the higher rate of investment in *physical* capital will also imply that more *human* capital per worker is accumulated. This reinforces the increase in output. Thus, the increase in GDP per person caused by a higher investment rate in physical capital is larger in a model including human capital than in one with only physical capital (other things, e.g. α , being equal).

The introduction of human capital should thus work to mitigate both of the empirical failures of the Solow model. How well a job it does in this respect is the subject of this chapter, and we will see that it does a very fine job.

6.1 The Solow model with human capital

The 'micro world' of the model to be presented is similar to that of the Solow models considered previously. There are the same kinds of economic agents, a representative profit-maximizing firm and consumers (government can again be interpreted into the model, a possibility that will be considered in an exercise). The production function of the firm is now different as it includes human capital, and the consumers will also have to decide on how large a fraction of income to accumulate as human capital each period. We assume that each unit of output can be used either as consumption, investment in physical capital, or as investment in human capital.

There are also the same competitive markets as before, one for output with real price normalized to 1, one for the services of physical capital with real rental rate, r_t , and one for labour with real wage rate, w_t . Hence we do not assume that there is a separate market for the services of human capital with a rental rate of its own. This is because human capital cannot be separated from workers. Its services have to be sold together with labour. So the services traded on the labour market will no longer be units of 'raw' labour, person-years, but person-years endowed with a certain level of human capital or education. This level is assumed to be the same for all workers. Hence the total stock of human capital, H_t , manifests itself as each worker in the labour force, L_t , holding a human capital of $h_t \equiv H_t/L_t$ that goes inseparably with the worker. Employing one more worker means hiring an additional unit of labour equipped with h_t .

The Solow model with human capital thus has two new features compared to the original Solow model: the inclusion of human capital in the production function and the accumulation of human capital by the consumers.

The production function with human capital

In period t the amounts of physical capital, K_t , and human capital, H_t , are predetermined by past accumulation. The aggregate production function is assumed to be:

$$Y_t = K_t^\alpha H_t^\varphi (A_t L_t)^{1-\alpha-\varphi}, \quad 0 < \alpha < 1, \quad 0 < \varphi < 1, \quad \alpha + \varphi < 1, \quad (1)$$

where A_t , the technological variable, is again assumed to follow an exogenous path given by a constant rate, g , of technological progress, where $g > -1$. Hence $A_t = A_0(1+g)^t$ for all t .

The production function (1) assumes constant returns to the three inputs K_t , H_t , and L_t . This accords with the replication argument. Given a technological level A_t , it should be possible to 'double production by doubling factor inputs', that is, by hiring $2L_t$ workers

instead of L_t , each being supplied with the same amount, $k_t \equiv K_t/L_t$, of physical capital, meaning that physical capital doubles to $2K_t$, and each also being endowed with the same amount, h_t , of human capital, so human capital doubles to $2H_t$. It follows that there are diminishing returns to the two reproducible capital inputs, $\alpha + \varphi < 1$.

Hiring one more (marginal) unit of labour now means hiring one more unit endowed with the average amount, h_t , of human capital per worker. Hence, a firm cannot increase the input of 'raw' labour, L_t , without increasing proportionally the input of human capital, $H_t = h_t L_t$. When we compute the marginal product of labour we should therefore consider h_t (not H_t) as being given. Inserting $H_t = h_t L_t$ into (1) gives:

$$Y_t = K_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{1-\alpha}. \quad (2)$$

Profit maximization and perfect competition in the markets for the services of physical capital and labour imply that the rental rates, r_t and w_t , are given by the marginal products of K_t and L_t (taking h_t as given in the computation of the latter):

$$r_t = \alpha K_t^{\alpha-1} h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{1-\alpha} = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{H_t}{A_t L_t} \right)^\varphi, \quad (3)$$

$$w_t = (1-\alpha) K_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{-\alpha} = (1-\alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{H_t}{A_t L_t} \right)^\varphi A_t. \quad (4)$$

From (3) and (4) as well as (2), it is easy to verify that the income shares of the owners of physical capital and of the workers are $r_t K_t / Y_t = \alpha$ and $w_t L_t / Y_t = 1 - \alpha$, respectively.

Dividing on both sides of (2) by L_t , and using the usual definition of output per worker, $y_t \equiv Y_t / L_t$, we get the 'per capita production function':

$$y_t = k_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi}, \quad (5)$$

from which, taking logs and time differences, and using the usual notation for the approximate growth rates, for example $g_t^y \equiv \ln y_t - \ln y_{t-1}$, one gets:

$$g_t^y = \alpha g_t^k + \varphi g_t^h + (1-\alpha-\varphi) g_t^A \approx \alpha g_t^k + \varphi g_t^h + (1-\alpha-\varphi) g_t. \quad (6)$$

In the Solow model with human capital, growth in GDP per worker can come from three sources: more physical capital per worker, more human capital per worker, and better technology. The (approximate) growth rate of y_t is the weighted average of the growth rates of k_t , h_t and A_t , the weights being the appropriate exponents in the Cobb–Douglas production function.

The empirical observation that the income share of workers is around $\frac{2}{3}$ gives an estimate of α of around $\frac{1}{3}$. Is there an empirical observation that could be used to give an impression of the magnitude of φ ? Mankiw, Romer and Weil suggest one in their paper. It may not give as firm a determination of φ as the relatively constant labour share does for α , but it does give some impression of the magnitude of φ . In US manufacturing the average wage has typically been around twice the minimum wage, perhaps somewhat larger. If we can interpret the minimum wage as the compensation for labour without education, and the average wage as the compensation for raw labour endowed with average education, this suggests that around one half of what a typical worker earns is really a compensation of raw labour, and the other half is a return to human capital. The income shares of physical capital, human capital and raw labour would then all be around $\frac{1}{3}$, with both of the two latter shares going to workers. Consider again the production function (1). It is evident that had we assumed that there was a separate market for the services of human capital with

marginal product compensation, etc., then the implied income share of human capital would have been ϕ . If the share of human capital of $\frac{1}{3}$ computed above can be equalized with this implied share, then ϕ should be around $\frac{1}{3}$. Therefore, a realistic calibration of parameters could be that both α and ϕ , and hence $1 - \alpha - \phi$, are around $\frac{1}{3}$.

Note that this calibration makes human capital as productive as physical capital in aggregate production. Omitting human capital may be a major source of error.

The accumulation of human capital

Each of the L_t households supplies one unit of labour inelastically in period t , and supplies all accumulated physical capital as long as the real rental rate, r_t , is positive. Again the number of households is assumed to change by an exogenous rate, n , where $n > -1$.

The representative consumer must decide how much to consume, C_t , and hence how much to save, $S_t = Y_t - C_t$, but this time also how to distribute total savings between gross investment in physical capital, I_t^K , and gross investment in human capital, I_t^H . Given these gross investments, the evolution in the capital stocks are:

$$K_{t+1} - K_t = I_t^K - \delta K_t, \quad 0 \ll 1 \quad (7)$$

$$H_{t+1} - H_t = I_t^H - \delta H_t, \quad 0 \ll 1 \quad (8)$$

where we have assumed that physical and human capital depreciate at the same rate. The intertemporal budget constraint that the choices of C_t (or S_t), I_t^K and I_t^H must obey, given Y_t is:

$$I_t^K + I_t^H = Y_t - C_t = S_t. \quad (9)$$

Again we do not consider any explicit dynamic optimization, but simply assume that the results of the consumers' decisions are that a given and constant fraction, s_K , of income is invested in physical capital:

$$I_t^K = s_K Y_t, \quad 0 < s_K < 1, \quad (10)$$

and a given fraction, s_H , is invested in human capital:

$$I_t^H = s_H Y_t, \quad 0 < s_H < 1. \quad (11)$$

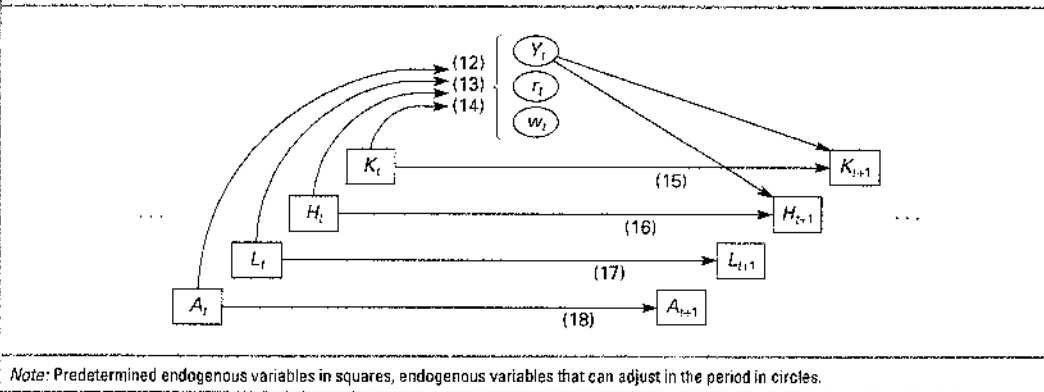
Hence from (9), $S_t = (s_K + s_H)Y_t$ and $C_t = (1 - s_K - s_H)Y_t$.² We assume $s_K + s_H < 1$.

The complete model

Since we have now described all the new model elements, we can write down the complete model consisting of the seven equations below. The first is the production function repeated from (1) above. The next two are the expressions for the factor prices, also derived above in (3) and (4). The following two are the capital accumulation equations that follow from (7) and (8) above with the assumed investment behaviour, (10) and (11), inserted. The last two equations contain the assumptions of given exogenous rates of change in labour force and technology, respectively.

2. The way we define consumption here does not correspond exactly to the usual national accounts definition where consumption is the excess of GDP over investment in *physical* capital only (and net exports).

FIGURE 6.1 The dynamics of the Solow model with human capital



THE COMPLETE SOLOW MODEL WITH HUMAN CAPITAL



$$Y_t = K_t^\alpha H_t^\varphi (A_t L_t)^{1-\alpha-\varphi}, \quad (12)$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{H_t}{A_t L_t} \right)^\varphi, \quad (13)$$

$$w_t = (1-\alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{H_t}{A_t L_t} \right)^\varphi A_t, \quad (14)$$

$$K_{t+1} - K_t = s_K Y_t - \delta K_t, \quad K_0 \text{ given} \quad (15)$$

$$H_{t+1} - H_t = s_H Y_t - \delta H_t, \quad H_0 \text{ given} \quad (16)$$

$$L_{t+1} = (1+n)L_t, \quad L_0 \text{ given} \quad (17)$$

$$A_{t+1} = (1+g)A_t, \quad A_0 \text{ given} \quad (18)$$

The parameters of the model are α , φ , s_K , s_H , δ , n , g . Given initial values, K_0 , H_0 , L_0 , A_0 , of the state variables, the model will determine the dynamic evolution of the economy as illustrated in Fig. 6.1.

The model is a *generalization* of the Solow model. If we set $\varphi = 0$, and consequently drop Eq. (16), the model of Chapter 5 appears (now denoting by s_K what was called s in Chapter 5). And if we also set $g = 0$, the basic Solow model of Chapter 3 appears. Just as in Chapter 5 we assume that $n + g + \delta + ng > 0$.

The law of motion

To analyse the Solow model with human capital it is again convenient to define 'technology adjusted' variables, physical capital per effective worker, $\tilde{k}_t \equiv k_t/A_t = K_t/(A_t L_t)$, human capital per effective worker, $\tilde{h}_t \equiv h_t/A_t = H_t/(A_t L_t)$, and output per effective worker, $\tilde{y}_t \equiv y_t/A_t = Y_t/(A_t L_t)$. Dividing both sides of (12) by $A_t L_t$ gives:

$$\tilde{y}_t = \tilde{k}_t^\alpha \tilde{h}_t^\varphi. \quad (19)$$

Writing the capital accumulation equations, (15) and (16), as $K_{t+1} = s_K Y_t + (1-\delta)K_t$ and $H_{t+1} = s_H Y_t + (1-\delta)H_t$, and dividing by $A_{t+1}L_{t+1}$ on both sides in each of them gives:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)}(s_K \tilde{y}_t + (1-\delta)\tilde{k}_t),$$

$$\tilde{h}_{t+1} = \frac{1}{(1+n)(1+g)}(s_H \tilde{y}_t + (1-\delta)\tilde{h}_t).$$

Inserting into these the expression for $\tilde{y}_t$ in (19) gives a law of motion for $\tilde{k}_t$ and $\tilde{h}_t$ consisting of the following two equations:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)}(s_K \tilde{k}_t^\alpha \tilde{h}_t^\phi + (1-\delta)\tilde{k}_t), \quad (20)$$

$$\tilde{h}_{t+1} = \frac{1}{(1+n)(1+g)}(s_H \tilde{k}_t^\alpha \tilde{h}_t^\phi + (1-\delta)\tilde{h}_t). \quad (21)$$

This law of motion is a bit more complicated than the transition equations you have seen before, since it consists of two coupled first-order difference equations (or a two-dimensional system of first-order difference equations). Given K_0, H_0, L_0, A_0 one can compute $\tilde{k}_0 = K_0/(A_0 L_0)$ and $\tilde{h}_0 = H_0/(A_0 L_0)$. Then $\tilde{k}_1$ can be computed from (20) and $\tilde{h}_1$ from (21). These two values can again be inserted into (20) and (21) to compute $\tilde{k}_2$ and $\tilde{h}_2$, etc. In this way the full sequences, $(\tilde{k}_t)$ and $(\tilde{h}_t)$, are determined. The new feature is that in each period t , the next $\tilde{k}_{t+1}$ depends on both $\tilde{k}_t$ and $\tilde{h}_t$, and likewise for $\tilde{h}_{t+1}$.

Given the sequences $(\tilde{k}_t)$ and $(\tilde{h}_t)$, the sequence $(\tilde{y}_t)$ follows from (19). The associated sequence (y_t) for GDP per worker follows from $y_t = \tilde{y}_t A_t$:

$$y_t = (1+g)^t A_0 \tilde{k}_t^\alpha \tilde{h}_t^\phi,$$

and the sequence (c_t) for consumption per worker follows from $c_t = (1 - s_K - s_H)y_t$, etc.

This time there is no simple diagrammatic way to demonstrate global convergence to a steady state. In fact, providing a global stability analysis of a two-dimensional system of *non-linear* difference equations is not easy. In the following we will limit ourselves to demonstrating convergence towards a steady state of the system (20) and (21) by doing a numerical simulation for realistic parameter values (and an exercise will ask you to do several more such simulations to check convergence), and by considering the linearization of the system around steady state to show convergence to steady state for the linearized system.

Before doing so we will *assume* stability and study the steady state in detail and develop a diagram that is useful for comparative analysis.

6.2 Steady state and comparative analysis in the Solow model with human capital

One can compute the steady state without first having established convergence to it over time. By definition a steady state is an economically meaningful ‘resting point’ for the dynamic system considered, in this case, two particular positive values for $\tilde{k}_t$ and $\tilde{h}_t$, which are such that if these values are ever reached, the economy stays at them forever.

The key endogenous variables in steady state

For the computation of steady state and for comparative static analysis it is convenient first to rewrite (20) and (21) as ‘Solow equations’ by subtracting $\tilde{k}_t$ from both sides of the first equation and $\tilde{h}_t$ from both sides of the second one:

$$\tilde{k}_{t+1} - \tilde{k}_t = \frac{1}{(1+n)(1+g)}(s_K \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n+g+\delta+ng)\tilde{k}_t), \quad (22)$$

$$\tilde{h}_{t+1} - \tilde{h}_t = \frac{1}{(1+n)(1+g)}(s_H \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n+g+\delta+ng)\tilde{h}_t). \quad (23)$$

A steady state is then a positive solution in $\tilde{k}_t$ and $\tilde{h}_t$ to the two equations that appear by setting the left-hand sides equal to zero:

$$s_K \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n+g+\delta+ng)\tilde{k}_t = 0, \quad (24)$$

$$s_H \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n+g+\delta+ng)\tilde{h}_t = 0. \quad (25)$$

Indicating the solution by an asterisk one has:

$$\tilde{k}^* = \left(\frac{s_K^{1-\varphi} s_H^\varphi}{n+g+\delta+ng} \right)^{\frac{1}{(1-\alpha-\varphi)}}, \quad (26)$$

$$\tilde{h}^* = \left(\frac{s_K^\alpha s_H^{1-\alpha}}{n+g+\delta+ng} \right)^{\frac{1}{(1-\alpha-\varphi)}}, \quad (27)$$

from which, also using (19), the steady state value for output per effective worker is:

$$\begin{aligned} \tilde{y}^* &= \left(\frac{s_K^{1-\varphi} s_H^\varphi}{n+g+\delta+ng} \right)^{\frac{\alpha}{(1-\alpha-\varphi)}} \left(\frac{s_K^{1-\alpha} s_H^\alpha}{n+g+\delta+ng} \right)^{\frac{\varphi}{(1-\alpha-\varphi)}} \\ &= \left(\frac{s_K}{n+g+\delta+ng} \right)^{\frac{\alpha}{(1-\alpha-\varphi)}} \left(\frac{s_H}{n+g+\delta+ng} \right)^{\frac{\varphi}{(1-\alpha-\varphi)}}. \end{aligned} \quad (28)$$

(As an exercise, do the derivations leading to the second equality.) We assume, realistically, that $n+g+\delta > 0$, so all expressions above and below are meaningful.

With the constant steady state values of the technology-adjusted variables in hand, we can find the steady state growth paths of the more interesting variables such as k_t , h_t and y_t . For instance, from $y_t = \tilde{y}_t A_t$, it follows that in steady state:

$$y_t^* = A_t \left(\frac{s_K}{n+g+\delta+ng} \right)^{\frac{\alpha}{(1-\alpha-\varphi)}} \left(\frac{s_H}{n+g+\delta+ng} \right)^{\frac{\varphi}{(1-\alpha-\varphi)}}, \quad (29)$$

where A_t is growing according to $A_0(1+g)^t$. Likewise, since $c_t = (1-s_K-s_H)y_t$, the growth path for consumption per worker in steady state is:

$$c_t^* = A_0(1+g)^t(1-s_K-s_H) \cdot \left(\frac{s_K}{n+g+\delta+ng} \right)^{\frac{\alpha}{(1-\alpha-\varphi)}} \left(\frac{s_H}{n+g+\delta+ng} \right)^{\frac{\varphi}{(1-\alpha-\varphi)}}. \quad (30)$$

Note from (29) that the exponent $\alpha/(1-\alpha-\varphi)$ is the elasticity of y_t^* with respect to s_K (show this), and this elasticity is larger than the $\alpha/(1-\alpha)$ of the general Solow model. This is as we wanted and it is due to the effect described in the introduction to this chapter. A higher s_K gives higher GDP per worker in steady state because of greater physical capital accumulation, but the increased GDP per worker now also generates more human capital per worker, reinforcing the increase in GDP per worker.

Notice also that the influence of n on y_t is now stronger than the influence of s_K (and of s_H), since n appears in both of the parentheses in (29). Disregarding the small product ng , the

elasticity of y_t with respect to $(n + g + \delta)$ is $-(\alpha + \varphi)/(1 - \alpha - \varphi)$ (show this), which is larger in absolute value than both of the elasticities appearing directly as exponents in (29). The reason is that increased population growth now gives a thinning out of *both* physical *and* human capital. This feature, that the rate of population growth affects GDP per worker more strongly than investment rates, is consistent with the empirical evidence, as you will be asked to verify in an exercise.

STEADY STATE



According to the steady state of the Solow model with human capital, the elasticity of long-run income per capita with respect to the investment rate s_K in physical capital is $\alpha/(1 - \alpha - \varphi)$, with respect to the investment rate s_H in human capital it is $\varphi/(1 - \alpha - \varphi)$, and with respect to $n + g + \delta$, it is $-(\alpha + \varphi)/(1 - \alpha - \varphi)$. The two first elasticities should both be around one, and the last one should be around minus two, given reasonable parameter values, $\alpha = \varphi = \frac{1}{3}$.

The phase diagram and comparative analysis

Consider again the two equations (24) and (25) from which we computed the steady state. The first of these gives the restriction on $\tilde{k}_t, \tilde{h}_t$ that follows from requiring $\tilde{k}_{t+1} - \tilde{k}_t = 0$, or in other words, it gives all the combinations of $\tilde{k}_t, \tilde{h}_t$ such that $\tilde{k}_t$ stays unchanged. We can draw these combinations in a $\tilde{k}_t, \tilde{h}_t$ diagram (see the curve labelled $[\Delta\tilde{k}_t = 0]$ in Fig. 6.2). You should check by isolating $\tilde{h}_t$ from (24) that indeed this curve should look like a $\tilde{k}_t^2$ function in the $\tilde{k}_t, \tilde{h}_t$ diagram for realistic parameter values (e.g., $\alpha = \varphi = \frac{1}{3}$). Likewise, (25) gives all the combinations of $\tilde{k}_t, \tilde{h}_t$ such that $\tilde{h}_t$ stays unchanged, and the curve for it, $[\Delta\tilde{h}_t = 0]$ in Fig. 6.2, should look like a $\sqrt{\tilde{k}_t}$ function (demonstrate this). The steady state combination, $\tilde{k}^*, \tilde{h}^*$, is where the two curves intersect.

It is also easy to see from (22) that for every combination of $\tilde{k}_t, \tilde{h}_t$ above the curve $[\Delta\tilde{k}_t = 0]$, one has $\tilde{k}_{t+1} - \tilde{k}_t > 0$, so in this area $\tilde{k}_t$ must be increasing, while below the same curve it must be decreasing as indicated by arrows. Likewise, everywhere to the right of the curve $[\Delta\tilde{h}_t = 0]$

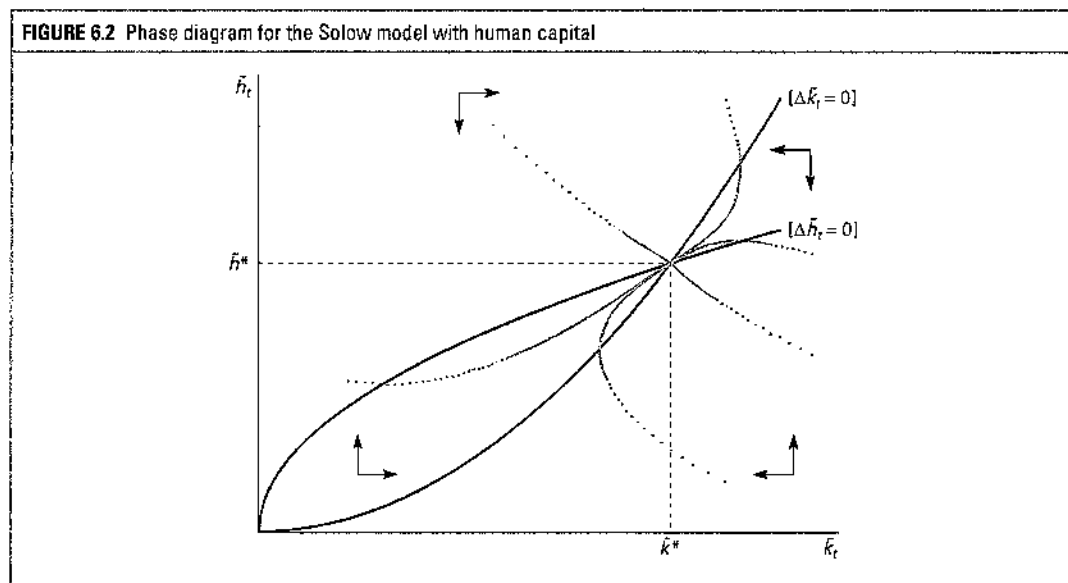
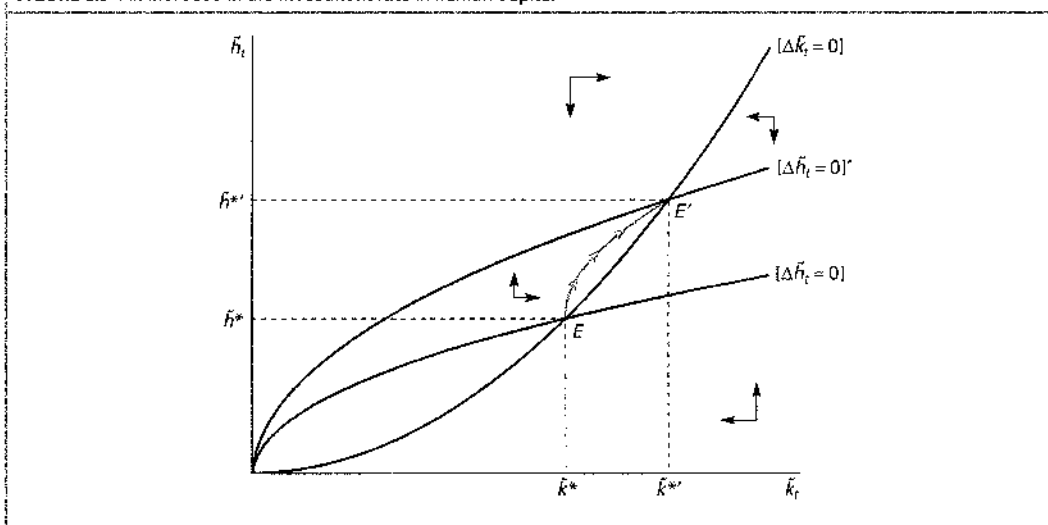


FIGURE 6.3 An increase in the investment rate in human capital



one has from (23) that $\dot{h}_{t+1} - \dot{h}_t > 0$, so in this area $\dot{h}_t$ increases over time, while to the left of the curve, $\dot{h}_t$ decreases as indicated by arrows.

From initial points $\bar{k}_0, \bar{h}_0$, like the alternative points shown in Fig. 6.2, $\bar{k}_t, \bar{h}_t$ will move along the trajectories as illustrated. These trajectories must respect the directions of change indicated by the arrows in the different regions. As Fig. 6.2 is drawn it suggests convergence to the steady state, but we emphasize that this does not follow just from the figure.

Figure 6.2 is called a phase diagram and it is useful for comparative analysis. Assume, for instance, that the economy is initially in steady state as illustrated by the intersection point E between the two curves $[\Delta \dot{k}_t = 0]$ and $[\Delta \dot{h}_t = 0]$ in Fig. 6.3. Suppose then that the investment rate in human capital, s_H , increases permanently, while the other parameters remain the same. You will easily show that this implies that the curve corresponding to unchanged $\dot{h}_t$ shifts upwards as illustrated by $[\Delta \dot{h}_t = 0]'$ in Fig. 6.3. As the phase diagram has now changed, the old steady state point E will be situated exactly on a boundary between two regions, and on this boundary $\dot{k}_t$ does not change while $\dot{h}_t$ increases. Once $\dot{h}_t$ has increased, the economy will be in the region where both $\dot{k}_t$ and $\dot{h}_t$ increase and there could be convergence to the new steady state E' along the trajectory indicated.³

What happens is that the increased s_H in the very beginning implies more accumulation of human capital (and only that), but as soon as the increased stock of human capital begins to generate increases in output, more physical capital will also be accumulated because of the constant rate of investment in physical capital. This explains why, along the indicated trajectory, both $\dot{k}_t$ and $\dot{h}_t$ increase.

During the transition to the new steady state both $\bar{k}_t$ and $\bar{h}_t$ increase, so $\bar{y}_t = \bar{k}_t^\alpha \bar{h}_t^{1-\alpha}$ must be increasing. Since $\bar{y}_t = y_t / A_t$, output per worker, y_t , must, during the transition, be increasing at a rate which is larger than g , but which falls back towards g as the new steady state is approached. The jump in the growth rate of y_t caused by the increase in s_H is thus much like the growth jump caused by an increased s_K in the general Solow model. An exercise will ask you to do simulations suggesting that responses to parameter changes do have the 'growth jump character'.

3. Note that we are assuming that the initial jump upwards in $\dot{h}_t$ is not so big that one ends in the upper region where $\dot{k}_t$ is increasing and $\dot{h}_t$ is decreasing. The (theoretical) possibility of such 'big jumps' is one reason why stability analysis is difficult here. Exercises will ask you to do numerical simulations that suggest that, for realistic parameter values, there will not be such big jumps, so the reaction to a parameter change is indeed as indicated by the grey curve in Fig. 6.3.

Empirics for steady state

To test the model's steady state prediction empirically, first take logs on both sides of (29) and use $ng \approx 0$:

$$\ln y_t^* \approx \ln A_t + \frac{\alpha}{1-\alpha-\varphi} [\ln s_K - \ln(n+g+\delta)] + \frac{\varphi}{1-\alpha-\varphi} [\ln s_H - \ln(n+g+\delta)]. \quad (31)$$

The relationship between $\ln y_t^*$ and the two explanatory variables, $[\ln s_K - \ln(n+g+\delta)]$ and $[\ln s_H - \ln(n+g+\delta)]$, is a linear one, and the relationship generalizes the one found in the general Solow model, the latter appearing by setting $\varphi = 0$. As already observed, if both α and φ are around $\frac{1}{3}$, then both the coefficients (or elasticities) in (31) should be around one.

With data for GDP per worker across countries i in year 2003 as well as data for s_K^i , s_H^i , and n^i , one will be able to estimate the regression equation:

$$\ln y_{03}^i = \gamma_0 + \gamma_1 [\ln s_K^i - \ln(n^i + 0.075)] + \gamma_2 [\ln s_H^i - \ln(n^i + 0.075)] + \xi^i, \quad (32)$$

suggested by the theoretical equation (31). The formulation of (32) assumes the same technological level A_{03} in year 2003 in all countries considered, the same technical parameters α and φ , as well as the same technological growth rate, g , and depreciation rate, δ . You may think these are heroic assumptions and we will discuss them further below.

We will consider the representative sample of 65 countries that we have used throughout Chapters 2, 3 and 5, and for which we have good data for y_{00}^i , s_K^i and n^i from the PWT 6.2. As before, we will for s_K^i use the average gross investment rate in physical capital between 1960 and 2003, and for n^i the average growth rate of the labour force between 1960 and 2003. But how can we get data for the investment rate, s_H , in human capital? This is required for an estimation of (32) and hence for testing the steady state prediction of our model.

In some countries most education costs are paid for by local and central public authorities, and for such countries one might consider measuring s_H as public expenditure on education divided by GDP. There are two problems with this. In many other countries much educational expenditure is paid privately, and in all countries the main part of the investment in education that consists of forgone income during education is paid privately. This leads to the idea of measuring s_H by a fraction that has as its denominator the GDP that *could* have been produced if all people of working age, including those in education, participated in production, and as its numerator, has the GDP that is *not* being produced because some people are in education. This fraction, the GDP lost due to education relative to potential GDP, would not include direct expenditure on education, but if this can be assumed to vary proportionally to lost GDP, one would have a good proxy.

It is not easy to get data for the fraction just described, but the fraction of people of working age in education to all people of working age should have a close relationship to it. The data available are such that even this latter fraction cannot be obtained directly. One *can*, however, get data across countries for the fraction of all people of school age (12–17) who attend secondary school. If we use this fraction as a proxy for the ‘degree of education’ among those who are of school age in a country, we can compensate for differences in the compositions of populations by multiplying it by the fraction of all people of working age who are of school age. For the latter we use the fraction of people between 15 and 19 to all people of working age (again because of available data). Our proxy for s_H is thus ‘the fraction of the school age population (12–17) attending secondary school times the fraction of the working age population that is of secondary school age (15–19)’. This is how the investment rates, s_H^i , reported in Table A across countries, i , are computed (averaged over the years 1960, 1965, . . . , 1995).⁴

4. Our figures for s_H are taken from the data set from Ben S. Bernanke and Refet S. Gürkaynak, ‘Is Growth Exogenous? Taking Mankiw, Romer, and Weil Seriously’, NBER Working Paper 8365, July 2001.

TABLE 6.1 Steady state regressions

Model:	Dependent variable: log of real GDP per worker, 2003 ($\ln y_{03}$)	
	Solow model without human capital	Solow model with human capital
Observations:	65	65
Explanatory Variables:		
Constant	8.92	9.82
$\ln s_K - \ln(n + 0.075)$	1.52 (0.14)	0.59 (0.18)
$\ln s_H - \ln(n + 0.075)$		0.93 (0.14)
Adjusted R^2	0.63	0.79
Implied values:		
α	0.60	0.23
φ	—	0.37

Note: Standard errors in parenthesis.

Source: Estimations based on Table A.

These rates are not correct measures with respect to their levels since they ignore direct expenditure on education, but if a correct measure could be assumed to be a constant times s_H^i , the constant being the same in all countries, then with the rates, s_H^i , as defined, we would arrive at the correct *relative* investment rates between countries. In the regression equation (32) the log of this constant times γ_2 would enter the intercept variable, γ_0 , by the same amount for all countries. Hence, an estimation of (32) based on the investment rates in human capital defined should give correct estimates of γ_1 and γ_2 .

The procedure described can deliver data for s_H^i for the 65 countries of our sample. An OLS estimation of (32) across these countries gives:

$$\ln y_{03}^i = 9.82 + \underset{(se=0.18)}{0.59} [\ln s_K^i - \ln(n^i + 0.075)] + \underset{(se=0.14)}{0.93} [\ln s_H^i - \ln(n^i + 0.075)], \quad (33)$$

$$\text{adj. } R^2 = 0.79,$$

where standard errors are indicated in parentheses below the estimates. This estimation is, in important respects, better than the corresponding one for the Solow model reported just after Fig. 5.7. For comparison we have included both regressions in Table 6.1 above, thereby also showing to you a way that multiple regressions are often presented.

The adjusted R^2 , indicating the fraction of the cross-country variation in GDP per worker explained by the equation, has increased from 0.63 to 0.79. It is indeed impressive that the simple equation we are considering should explain 79 per cent of the per capita income variation in the world. Yet, the increase in R^2 is not the main reason for considering the present equation and estimation an improved one.

The main reason is that the sizes of the estimated parameters, the 0.59 and 0.93, respectively, are now in much better accordance with the values we would expect based on the empirical observations that suggest an α of around $\frac{1}{3}$, and (with less precision) a φ also around $\frac{1}{3}$. As mentioned, this implies a hypothesis that the coefficients, γ_1 and γ_2 , are of equal size and both around one. Our estimates of γ_1 and γ_2 are not exactly equal and only the estimate of γ_2 is truly close to one, but taking uncertainty into account, one cannot reject the hypothesis considered. The 95 per cent confidence intervals for γ_1 and γ_2 (each of which goes approximately from two standard deviations below to two standard deviations above

the estimate) overlap and the interval for γ_1 reaches up close to one.⁵ Note that a similar accordance between theoretical and estimated values cannot be obtained for the γ of the Solow model considered in Chapter 5. The estimate was 1.52 with the 95 per cent confidence interval going down to 1.23, which is still far from the theoretical value for γ of 0.5.

Another way to illustrate the better accordance with the theory is to compute the estimates of α and ϕ implied by the estimates of γ_1 and γ_2 . Since γ_1 should be $\alpha/(1 - \alpha - \phi)$, and γ_2 should be $\phi/(1 - \alpha - \phi)$, we have two equations in two unknowns that can easily be solved to find:

$$\alpha = \frac{\gamma_1}{1 + \gamma_1 + \gamma_2} \quad \text{and} \quad \phi = \frac{\gamma_2}{1 + \gamma_1 + \gamma_2}.$$

Inserting the estimated values for γ_1 and γ_2 gives, as also reported in Table 6.1, $\alpha = 0.23$ and $\phi = 0.37$, which are not too far from the 'correct' values of *around* $\frac{1}{3}$ and $\frac{1}{3}$.

We can also give a graphical illustration of the improvement in our model obtained by including human capital. Consider first the steady state equation of the Solow model *without* human capital arising by setting $\phi = 0$ (still using $ng \equiv 0$):

$$\ln y_i^* = \ln A_i + \frac{\alpha}{1 - \alpha} [\ln s_K - \ln(n + g + \delta)].$$

We tested this in Chapter 5 by plotting $\ln y_{00}^i$ against $\ln s_K^i - \ln(n^i + 0.075)$, finding a nice linear relationship, but not with a slope of 0.5 as it should be. We could also have chosen to plot $\ln y_{00}^i$ against $0.5[\ln s_K^i - \ln(n^i + 0.075)]$, the difference being the addition of the factor 0.5. Assuming a value of $\frac{1}{3}$ for capital's share, α , the plotted points should be close to a straight line *with a slope of one* for the model without human capital to be correct. Fig. 6.4 shows the plot for the 65 countries of our sample. The grey line is an OLS regression line through the observations. The black one is a straight line with the same intercept as the grey one and with a slope of one. The figure shows that the data points may be situated along a straight line, but not along one with a slope of one.

Doing the same kind of graphical test for the Solow model *with* human capital consists of plotting $\ln y_{00}^i$ against $[\ln s_K^i - \ln(n^i + 0.075)] + [\ln s_H^i - \ln(n^i + 0.075)]$, where we have set both coefficients at one, which follows from $\alpha = \phi = \frac{1}{3}$. The result is shown in Fig. 6.5. Compared to Fig. 6.4, the points are now clustered around a grey OLS line that has a slope that is relatively close to one (like the black line). To conclude:

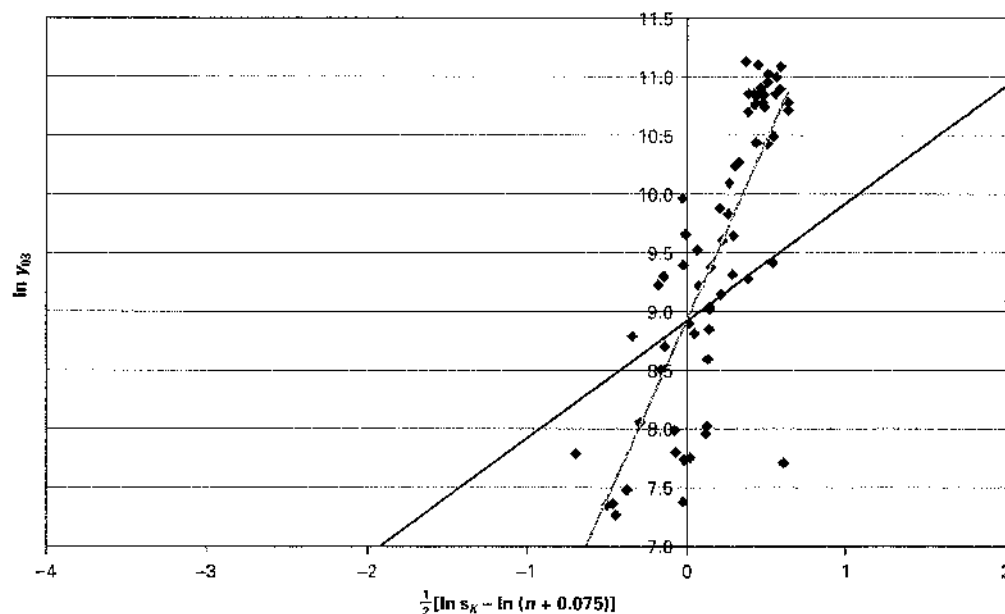
STEADY STATE AND THE CROSS-COUNTRY EMPIRICAL EVIDENCE



The steady state prediction of the Solow model with human capital accords remarkably well with cross-country empirical evidence even under an assumption of a common technological level. We have explained about four-fifths of the cross-country variation in GDP per worker using a few explanatory variables, s_K , s_H and n , which have been constrained to appear in our regression equation as required by our theoretical model.

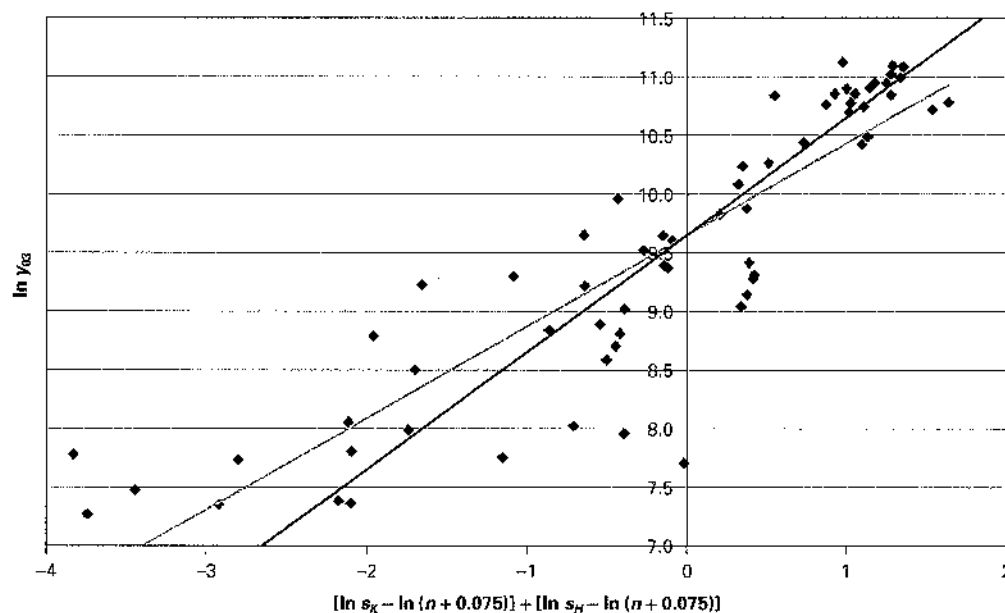
The good fit of Fig. 6.5 does not *prove* that the Solow model with human capital is correct. As warned also in Chapter 5, the problem of 'reversed causality' could be relevant here: is it really s_K , s_H and n that explain y , as the model assumes, or is it rather a high y that generates high values of s_K and s_H , and a low n ? Or Fig. 6.5 could be a reflection of 'spurious

5. By including a bit of uncertainty with respect to the true values of α and ϕ , the accordance can be made even better. Based on empirical observations we can only say that α and ϕ should be *around* $\frac{1}{3}$. If they were both 0.3 say (rather than 0.333 . . .), then both of γ_1 and γ_2 should be 0.75, which is in both 95 per cent confidence intervals.

FIGURE 6.4 Logarithm of real GDP per worker in 2003 against structural variables, s_K and n , 85 countries


Note: The estimated line is $y = 8.92 + 3.04x$, and has $R^2 = 0.64$. The standard error of the estimate of the slope is 0.29.

Source: Penn World Table 6.2.

FIGURE 6.5 Logarithm of real GDP per worker in 2003 against structural variables, s_K , s_H , and n , 85 countries


Note: The estimated line is $y = 9.66 + 0.78x$, and has $R^2 = 0.79$. The standard error of the estimate of the slope is 0.05.

Source: Penn World Table 6.2. Investment rates in human capital from Bernanke and Gürkaynak (2001).

correlation' where it is really omitted factors that cause s_K , s_H , n and y to move in a way that creates the correlation observed. Nevertheless, it is certainly reassuring for the model that its prediction fits the data so well.

The estimations and discussion above assume a common technological level, A_{03} , in 2003 among the countries considered. It is an important detail that this assumption does not seem to be strongly contradicted by the evidence: the points in Fig. 6.5 *do* seem to be situated along one and the same straight line with a slope close to one and this *is* evidence in favour of a common A_{03} . It may be surprising that in explaining the income differences in the world we do not necessarily have to rely strongly on 'technological differences', but can, to a large degree, see income differences as reflections of differences in investment rates and labour force growth rates. In other words, the income differences seem to a large extent to be explained by differences in the *amounts* of inputs (physical and human capital per worker), whereas differences in the *productivities* of these inputs appear to be less important. A rough way to state this conclusion (or hypothesis) is that computers and education in poorer countries are as advanced as in richer countries, there are just fewer computers and less education per capita in the poorer countries because of lower investment rates and higher population growth rates. In Section 6.4 below we return to this issue and investigate further how important 'technological differences' are in understanding the difference between rich and poor.

Structural policy for steady state

Since the model's steady state prediction fits as well as it does with the evidence, one can feel somewhat safe in using the model for recommendations concerning long-run structural policies. With respect to policies intended to affect investment rates in physical capital and population growth rates, the conclusions go in the same direction as in previous chapters. The present model may be more correct with respect to the sizes of the underlying elasticities, which is important for policy, but qualitatively the policy implications are not affected. It is worth repeating, however, that according to the present model, the elasticity of GDP per person with respect to population growth (or rather to $n + g + \delta$) is larger in absolute value than each of the elasticities with respect to the investment rates, s_K and s_H .

It is with respect to investment in human capital that the model may have new policy implications. First, note that according to (29), steady state GDP per worker is unambiguously increasing in the investment rate s_H , and according to the model as well as to the empirics investment in human capital seems to be at least as important for GDP per worker as investment in physical capital. This means that more investment in education may have desirable effects, but not under all circumstances. First, given the model's other parameters, the 'golden rule' values for s_K and s_H that will maximize the altitude of the growth path for consumption per worker are $s_K = \alpha$ and $s_H = \varphi$ (an exercise will ask you to show this). Raising the investment rates above these values is not meaningful.

Second, a public policy to provide for or subsidize education should be justified by specific reasons why leaving educational decisions to private individuals cannot be expected to yield optimal outcomes. In the case of education there are many such reasons. One is imperfect credit markets. If education were a completely private matter, people who would like to educate themselves would have to borrow money against their expected higher future labour incomes to provide for the costs of education and to compensate for the forgone income during education. However, there are strict limits to how much private banks will lend to a (poor) person based on this person's expected future income, for instance due to problems of 'adverse selection': a bank would probably be perfectly ready to lend to the able and honest people who will actually come to earn more and be able and willing to pay back after education, but not to other people. It may, however, be difficult for the bank to distinguish between the two types of people and therefore it will be reluctant to lend for educational purposes at all.

Another reason is imperfect insurance markets. More education does not yield a specific increase in future income with certainty, but rather with some probability. A decision on education may fail, so connected to education is an insurance problem. Few private insurance companies are prepared to sell insurance policies that pay out money in cases of failed education, this time due to 'moral hazard' problems: a person who is well insured against the event of failed education will not have so strong an incentive to work hard for making education a success since there will be payout under all circumstances. Therefore insurance companies will be reluctant to insure against failed education.

There are probably also strong positive externalities associated with education. Part of the return on an investment in education goes to other people. For instance, an engineer in a private firm is more productive if the other employees in the firm can read and write and do calculations. This means that part of the social return to the education invested in these employees appears as higher productivity of the engineer and is not appropriated by those who acquired the education. If left to private decisions, people will therefore probably educate themselves too little as compared to a social optimum.

The microeconomic arguments in favour of subsidizing education that we have presented are not part of the human capital growth model studied in this chapter, but the strong effects of investment in education that the model and the empirics point to indicate that these arguments are not just of academic interest. They are probably of interest quantitatively as well.

In any case, the issues of how and to what extent to provide for and subsidize education are among the most important concerns of public policy.

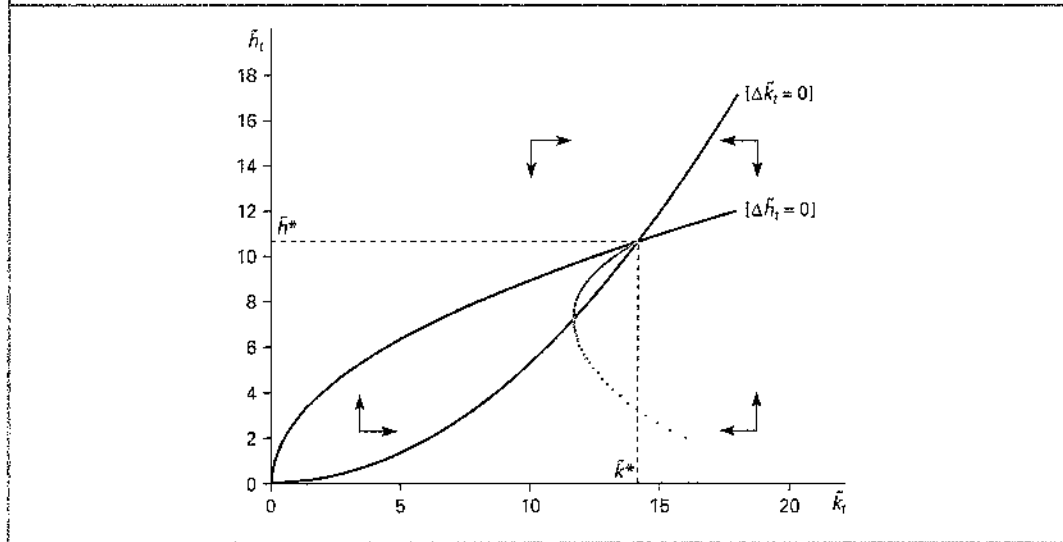
6.3 Convergence in the Solow model with human capital⁶

The steady state of the Solow model with human capital accords in all respects with positive balanced growth if $g > 0$. It is left to you as an exercise to demonstrate this in detail, but note that since $\bar{k}_t$, $\bar{h}_t$ and $\bar{y}_t$ are all locked at constant values in steady state, $\dot{k}_t$, $\dot{h}_t$ and $\dot{y}_t$ must all be increasing at the same rate as A_t , that is, at rate g .

The Solow model with human capital thus passes the 'long-run empirical growth check', but this time we have not shown that the steady state equilibrium is stable. Taking stability for granted (for the time being), as the economy converges to steady state there will be transitory growth in addition to the underlying technological growth, and the transitory growth seems to be more complicated than in the general Solow model since there are now two state variables, $\bar{k}_t$ and $\bar{h}_t$, in the dynamic system. For instance, outside steady state one variable could be below and the other one above its steady state value.

By conducting a numerical simulation exercise with the system (20) and (21), we will start by indicating that this system *does* imply convergence to steady state. We will then study the linear approximation of the system around steady state. Although this is a *two-dimensional* linear difference equation in $\bar{k}_t$ and $\bar{h}_t$, it turns out that it implies a one-dimensional linear difference equation in $\bar{y}_t$, which has the same form as the corresponding equation for the general Solow model. This has three implications. First, the convergence process in terms of $\bar{y}_t$ is like the one we have studied for the general Solow model: the model with human capital also supports conditional convergence. Second, we can show analytically that the linearized system is stable towards steady state. Third, we can test the convergence equation empirically and we will find that the rate of convergence predicted by the model is now in much better accordance with empirical evidence than was the case for the general Solow model.

6. Some readers may find this section difficult and it is possible to understand the remainder of the chapter and the book without reading it. Until the subsection 'The Convergence Process . . .' there should be no difficulties, however.

FIGURE 6.6 Simulation of convergence in the Solow model with human capital

Illustrating convergence to steady state by simulation

Let us specify some plausible parameter values: $\alpha = \varphi = \frac{1}{3}$, $s_K = 0.2$, $s_H = 0.15$, $\delta = 0.06$, $n = 0$ and $g = 0.015$. The steady state values $\bar{k}^*$ and $\bar{h}^*$ are easily computed from (26) and (27): $\bar{k}^* = 14.22$ and $\bar{h}^* = 10.67$. We consider a starting point where one of the state variables is above its steady state value and the other below: $\bar{k}_0 = 16$ and $\bar{h}_0 = 2$. Starting from these values we simulate the system (20) and (21) with the chosen parameter values. The result is shown in Fig. 6.6, a $\bar{k}_t, \bar{h}_t$ diagram where each dot corresponds to one period. The figure also has the two black curves $[\Delta \bar{k}_t = 0]$ and $[\Delta \bar{h}_t = 0]$, which makes it a phase diagram.

The sequence of dots in Fig. 6.6 converges to the steady state. An exercise will ask you to do simulations from other starting points and using other parameter combinations. You should find convergence to steady state. To construct Fig. 6.2 we actually simulated the model with alternative starting points and the same parameter values as considered here, so Fig. 6.2 contributes to the impression of general convergence and it suggests the type of picture you are supposed to find when you attempt the exercise.

CONVERGENCE TO STEADY STATE



The Solow model with human capital implies for reasonable parameter values convergence to the model's steady state. The steady state fulfils the requirements for balanced growth in all respects (Exercise 3).

The convergence process according to the Solow model with human capital

To obtain a linear approximation of the system (20) and (21), first write it in the generic form:

$$\bar{k}_{t+1} = G(\bar{k}_t, \bar{h}_t),$$

$$\bar{h}_{t+1} = L(\bar{k}_t, \bar{h}_t),$$

where the functions G and L are defined as the right-hand sides of (20) and (21), respectively. Linearizing around steady state turns the first of these equations into:

$$\tilde{k}_{t+1} - \tilde{k}^* = G_k(\tilde{k}^*, \tilde{h}^*)(\tilde{k}_t - \tilde{k}^*) + G_h(\tilde{k}^*, \tilde{h}^*)(\tilde{h}_t - \tilde{h}^*),$$

where G_k and G_h are the partial derivatives of G with respect to $\tilde{k}_t$ and $\tilde{h}_t$, respectively, and similarly for the other equation. If we use $\tilde{k}_{t+1} - \tilde{k}^* \approx \tilde{k}^*(\ln \tilde{k}_{t+1} - \ln \tilde{k}^*)$ etc. (just as in the previous chapter), we get that, *up to a linear approximation around steady state*, the system (20) and (21) is equivalent to:

$$\ln \tilde{k}_{t+1} - \ln \tilde{k}^* = G_k(\tilde{k}^*, \tilde{h}^*)(\ln \tilde{k}_t - \ln \tilde{k}^*) + \frac{\tilde{h}^*}{\tilde{k}^*} G_h(\tilde{k}^*, \tilde{h}^*)(\ln \tilde{h}_t - \ln \tilde{h}^*), \quad (34)$$

$$\ln \tilde{h}_{t+1} - \ln \tilde{h}^* = \frac{\tilde{k}^*}{\tilde{h}^*} L_k(\tilde{k}^*, \tilde{h}^*)(\ln \tilde{k}_t - \ln \tilde{k}^*) + L_h(\tilde{k}^*, \tilde{h}^*)(\ln \tilde{h}_t - \ln \tilde{h}^*). \quad (35)$$

This is a two-dimensional *linear* system of difference equations in the relative discrepancies (from steady state): $\ln \tilde{k}_t - \ln \tilde{k}^*$ and $\ln \tilde{h}_t - \ln \tilde{h}^*$. The coefficients in the system can be computed from the partial derivatives of G and L . Doing the differentiations and using $\tilde{y}^* = (\tilde{k}^*)^\alpha (\tilde{h}^*)^\varphi$ gives:

$$G_k(\tilde{k}^*, \tilde{h}^*) = R \left(s_K \alpha \frac{\tilde{y}^*}{\tilde{k}^*} + (1 - \delta) \right), \quad \frac{\tilde{h}^*}{\tilde{k}^*} G_h(\tilde{k}^*, \tilde{h}^*) = R s_K \varphi \frac{\tilde{y}^*}{\tilde{k}^*},$$

$$\frac{\tilde{k}^*}{\tilde{h}^*} L_k(\tilde{k}^*, \tilde{h}^*) = R s_H \alpha \frac{\tilde{y}^*}{\tilde{h}^*}, \quad L_h(\tilde{k}^*, \tilde{h}^*) = R \left(s_H \varphi \frac{\tilde{y}^*}{\tilde{h}^*} + (1 - \delta) \right),$$

where we have used R to denote $1/[(1+n)(1+g)]$. From the steady state expressions (26) to (28) we get:

$$\frac{\tilde{y}^*}{\tilde{k}^*} = \frac{n+g+\delta+ng}{s_K},$$

$$\frac{\tilde{y}^*}{\tilde{h}^*} = \frac{n+g+\delta+ng}{s_H},$$

$$\left(\frac{\tilde{k}^*}{\tilde{h}^*} = \frac{s_K}{s_H} \right).$$

Inserting into (34) and (35) then gives the linearized system:

$$\ln \tilde{k}_{t+1} - \ln \tilde{k}^* = R[\alpha(n+g+\delta+ng) + (1-\delta)](\ln \tilde{k}_t - \ln \tilde{k}^*) + R\varphi(n+g+\delta+ng)(\ln \tilde{h}_t - \ln \tilde{h}^*),$$

$$\ln \tilde{h}_{t+1} - \ln \tilde{h}^* = R\alpha(n+g+\delta+ng)(\ln \tilde{k}_t - \ln \tilde{k}^*) + R[\varphi(n+g+\delta+ng) + (1-\delta)](\ln \tilde{h}_t - \ln \tilde{h}^*).$$

It turns out that this system implies a specific one-dimensional and linear difference equation in $\ln \tilde{y}_t - \ln \tilde{y}^*$, to which we will limit our attention. From $\tilde{y}_t = \tilde{k}_t^\alpha \tilde{h}_t^\varphi$, it follows that in any period, $\ln \tilde{y}_t = \alpha \ln \tilde{k}_t + \varphi \ln \tilde{h}_t$. Hence, multiplying the first of the above two dynamic equations by α , and the second by φ , and then adding the two gives:

$$\begin{aligned} \ln \tilde{y}_{t+1} - \ln \tilde{y}^* = & R[(\alpha + \varphi)(n+g+\delta+ng) + (1-\delta)]\alpha(\ln \tilde{k}_t - \ln \tilde{k}^*) \\ & + R[(\alpha + \varphi)(n+g+\delta+ng) + (1-\delta)]\varphi(\ln \tilde{h}_t - \ln \tilde{h}^*), \end{aligned}$$

and then, moving the common factor on the right-hand side outside the bracket we get:

$$\ln \tilde{y}_{t+1} - \ln \tilde{y}^* = R[(\alpha + \varphi)(n + g + \delta + ng) + (1 - \delta)](\ln \tilde{y}_t - \ln \tilde{y}^*). \quad (36)$$

This is a first-order linear difference equation in $\ln \tilde{y}_t - \ln \tilde{y}^*$. The coefficient on the right-hand side is positive and you will easily (and should) verify that it is smaller than one if $\alpha + \varphi < 1$ and $n + g + \delta + ng > 0$, which are among our assumptions. Hence (36) implies stability, so according to it $\tilde{y}_t \rightarrow \tilde{y}^*$ as $t \rightarrow \infty$. You will recognize that (36) has exactly the same structure as (33) in Chapter 5.

Subtracting $\ln \tilde{y}_t$ and adding $\ln \tilde{y}^*$ on both sides of (36), and after that inserting for R gives:

$$\begin{aligned} \ln \tilde{y}_{t+1} - \ln \tilde{y}_t &= [1 - R[(\alpha + \varphi)(n + g + \delta + ng) + (1 - \delta)]](\ln \tilde{y}^* - \ln \tilde{y}_t) \\ &= \frac{(1+n)(1+g) - (\alpha + \varphi)(n + g + \delta + ng) - (1 - \delta)}{(1+n)(1+g)} (\ln \tilde{y}^* - \ln \tilde{y}_t), \end{aligned}$$

which, using $(1+n)(1+g) - (1 - \delta) = n + g + \delta + ng$ (check) gives:

$$\ln \tilde{y}_{t+1} - \ln \tilde{y}_t = \frac{(1 - \alpha - \varphi)(n + g + \delta + ng)}{(1+n)(1+g)} (\ln \tilde{y}^* - \ln \tilde{y}_t). \quad (37)$$

Just as in Chapter 5, we have arrived at a dynamic equation of the form $\ln \tilde{y}_{t+1} - \ln \tilde{y}_t = \lambda(\ln \tilde{y}^* - \ln \tilde{y}_t)$, now with the rate of convergence:

$$\lambda = \frac{(1 - \alpha - \varphi)(n + g + \delta + ng)}{(1+n)(1+g)} \approx (1 - \alpha - \varphi)(n + g + \delta), \quad (38)$$

where $0 < \lambda < 1$ (since λ is one minus the coefficient we found above to be between zero and one). Consequently our dynamic equation, (37), has the convergence property, that the further below steady state the economy is, the larger is the growth rate of $\tilde{y}_t$, and hence of y_t . The model thus supports conditional convergence.

In Chapter 5 we found the solution to the dynamic equation, $\ln \tilde{y}_{t+1} - \ln \tilde{y}_t = \lambda(\ln \tilde{y}^* - \ln \tilde{y}_t)$, after having written it in the standard form, $\ln \tilde{y}_{t+1} = \lambda \ln \tilde{y}^* + (1 - \lambda) \ln \tilde{y}_t$ (equation (36) of Chapter 5), and applied a cookbook solution procedure. The solution we found was (38) of Chapter 5, repeated here:

$$\frac{\ln y_T - \ln y_0}{T} = \frac{\ln A_T - \ln A_0}{T} + \frac{1 - (1 - \lambda)^T}{T} (\ln A_0 + \ln \tilde{y}^* - \ln y_0).$$

Inserting the steady state value $\tilde{y}^*$ from (29) and $(\ln A_T - \ln A_0)/T \approx g$ we get:

$$\begin{aligned} \frac{\ln y_T - \ln y_0}{T} &\approx g + \frac{1 - (1 - \lambda)^T}{T} \ln A_0 - \frac{1 - (1 - \lambda)^T}{T} \ln y_0 \\ &\quad + \frac{1 - (1 - \lambda)^T}{T} \frac{\alpha}{1 - \alpha - \varphi} [\ln s_K - \ln(n + g + \delta + ng)] \\ &\quad + \frac{1 - (1 - \lambda)^T}{T} \frac{\varphi}{1 - \alpha - \varphi} [\ln s_H - \ln(n + g + \delta + ng)]. \end{aligned} \quad (39)$$

This is the convergence equation for the Solow model with human capital. It tells how growth should depend on the structural characteristics and on the initial position. Note from (38) that we have obtained a lower rate of convergence than for the Solow model without human capital, $(1 - \alpha - \varphi)(n + g + \delta)$ instead of $(1 - \alpha)(n + g + \delta)$. Realistic parameter

values, e.g. $n + g + \delta$ between 0.06 to 0.09, and α and ϕ both around $\frac{1}{3}$, now imply a rate of convergence between 2 and 3 per cent, which should be compared to the 4 to 6 per cent from the general Solow model. The model-based rate of convergence has thus come much closer to our earlier empirical estimates, but these were not rooted in the present model. We should now do an estimation based on the new convergence equation (39) and see how well the equation describes the data and how well the estimate of λ , implied by the estimation, fits with the model-based values. We will do this shortly, but first sum up:

CONVERGENCE THEORETICALLY



Convergence property: Income per worker converges towards a steady state growth path along which the growth rate is g , and which is fully characterized by the structural variables, s_K , s_H , n , g , δ , and A_0 . The growth rate of income per worker is higher the lower the current level of income per worker.

Speed of convergence: The rate of convergence (to steady state), λ , is approximately $(1 - \alpha - \phi)(n + g + \delta)$, which on an annual basis should be in the range of 2 to 3 per cent for plausible parameter values.

Testing the convergence equation and estimating the rate of convergence

The theoretical equation (39) suggests a regression equation across countries i :

$$g_{T,0}^i = \beta_0 - \beta_1 \ln y_0^i + \beta_2 [\ln s_K^i - \ln(n^i + 0.075)] + \beta_3 [\ln s_H^i - \ln(n^i + 0.075)] + \xi^i, \quad (40)$$

where $g_{T,0}^i$ is the observed growth rate of GDP per worker from year 0 to year T . The coefficients β_1 , β_2 and β_3 should be linked to the parameters of the model as given by (39). Moreover, β_0 should depend on the underlying rate of technological progress, g , and the initial level, A_0 , as in (39). Hence, (40) assumes this latter technological influence on growth to be the same for all countries.

An OLS estimation of (40) based on the 65 countries of our representative sample gives:

$$g_{03,60}^i = 0.097 - 0.009 \ln y_0^i + 0.011 [\ln s_K^i - \ln(n^i + 0.075)] + 0.008 [\ln s_H^i - \ln(n^i + 0.075)],$$

(se=0.002) (se=0.003) (se=0.003)

adj. $R^2 = 0.45$. (41)

For comparison Table 6.2 includes the corresponding growth regression of Chapter 5 based on the Solow model without human capital, the estimation in (41), plus one where the explanatory variables, $\ln s_K^i$, $\ln s_H^i$, and $\ln(n^i + 0.075)$ have not been restricted to enter exactly as the model says, but appear separately and linearly on the right-hand side.

There are several nice features about the estimation in (41). First note that the adjusted R^2 is 0.45, so apparently the simple equation explains almost half of the cross-country variation in growth rates. This is pretty good, and better than the 40 per cent we arrived at in the general Solow model. However, the increase in R^2 is not the main reason for considering this estimation a much improved one. The main reason is that the estimated coefficients are now in better accordance with the underlying model.

First, note that according to the theory, still assuming that α and ϕ are of equal size, β_2 and β_3 should also be of equal size and their estimates, the two coefficients on the square bracket terms in (41), are indeed roughly equal taking uncertainty into account. Furthermore, we can derive the magnitudes of λ , α , and ϕ from the estimation. The parameter β_1 should be $(1 - (1 - \lambda)^T)T$, implying (as in Chapter 5) $\lambda = 1 - (1 - T\beta_1)^{1/T}$. Inserting $T = 43$, and $\beta_1 = 0.009$, gives $\lambda = 0.011$. Hence, our estimate of the (annual) rate of convergence is now 1.1 per cent

TABLE 6.2 Growth regressions

Dependent variable: average annual growth rate of real GDP per worker, 1960–2003, (g^*)			
Model:	Solow without human capital	Solow with human capital, restricted	Solow with human capital, unrestricted
Observations:	65	65	65
Explanatory Variables:			
Constant	0.062	0.095	0.039
$\ln y_0$	-0.006 (0.001)	-0.009 (0.002)	-0.01 (0.002)
$\ln s_K - \ln(n + 0.075)$	0.016 (0.002)	0.011 (0.003)	
$\ln s_H - \ln(n + 0.075)$		0.008 (0.003)	
$\ln s_K$			0.009 (0.003)
$\ln s_H$			0.009 (0.003)
$\ln(n + 0.075)$			-0.045 (0.013)
Adjusted R^2	0.38	0.45	0.48
Implied values:			
α	0.73	0.39	—
ϕ	—	0.30	—
λ	0.007	0.011	—

Note: Standard errors in parenthesis.

Source: Estimations based on Table A.

as also reported in Table 6.2. The 95 per cent confidence interval for λ ranges from 0.6 per cent up to 1.8 per cent (compute this yourself). Our estimate of the rate of convergence is clearly in better accordance with the ‘theoretical’ value arising from the present model’s rate of convergence, 2 to 3 per cent, than was the case for the Chapter 5 regression where the estimated λ of 0.7 per cent (with confidence interval from 0.3 to 0.9 per cent) did not accord well with the ‘theoretical’ 4 to 6 per cent arising from that model.

It is left to you as an exercise to find a way to derive α and ϕ from our estimation, and to find the implied magnitudes, but the solution is $\alpha = 0.39$ and $\phi = 0.30$. These values are in nice accordance with the expected values based on empirical observations, both around $\frac{1}{3}$.

The unrestricted estimation reported in the last column in Table 6.2 confirms the restricted one in some respects: the estimates of the coefficients on $\ln y_0$, $\ln s_K$, and $\ln s_H$ are very close to their counterparts for the restricted estimation. However, the coefficient on $\ln(n + 0.075)$ is estimated to be -0.045, which is of the right sign, but five times larger in absolute value than the coefficients on $\ln s_K$ and $\ln s_H$, where, according to the model, it should only have been twice as large. This is a point where the unrestricted estimation seems not to confirm our model. The next chapter, which incorporates scarce natural resources into the Solow model, may contribute to an explanation why growth in the labour force has a stronger negative influence on economic growth than predicted by the present model.

CONVERGENCE EMPIRICALLY



The out-of-steady-state convergence prediction of the Solow model with human capital fits the cross-country data remarkably well: controlling for structural influences there is a clear negative relationship between the initial value of and the subsequent growth in income per worker across the countries of the world, and the estimated speed of convergence is now in better accordance with the theoretical prediction of the model than was the case for the Solow model.

Figure 6.7 illustrates the gradual improvement of the explanation of growth and convergence we have achieved. All three panels include the 65 countries of our sample. The top panel tests (again) unconditional convergence and simply plots average annual growth in GDP per worker from 1960 to 2003 against the log of GDP per worker in 1960. It is thus identical to Fig. 2.5 and shows no tendency that the initially poorest countries should grow fastest *unconditionally*.

The middle panel illustrates conditional convergence according to the general Solow model. It is close to, but not the same as, Figs 2.6 and 5.8. Based on our estimation of the convergence equation of the general Solow model, Fig. 5.8 plotted the 'growth rate controlled for structural characteristics' $g_{03,60}^i - 0.016[\ln s^i - \ln(n^i + 0.075)]$ against $\ln y_{60}^i$. The middle panel in Fig. 6.7 does almost the same, but plots the 'growth rate controlled for structural characteristics' (defined the same way) *plus the average of structural influence over all the countries* against $\ln y_{60}^i$. The difference lies in the addition of the average, which makes the scales along the vertical axes in the panels of Fig. 6.7 comparable.

The bottom panel does for the Solow model with human capital what the middle panel does for the general Solow model, that is, it plots the growth rate $g_{03,60}^i$ minus the structural influence, $0.011[\ln s_k^i - \ln(n^i + 0.075)] + 0.008[\ln s_H^i - \ln(n^i + 0.075)]$, plus the average of these structural influences over all the countries against $\ln y_{60}^i$.

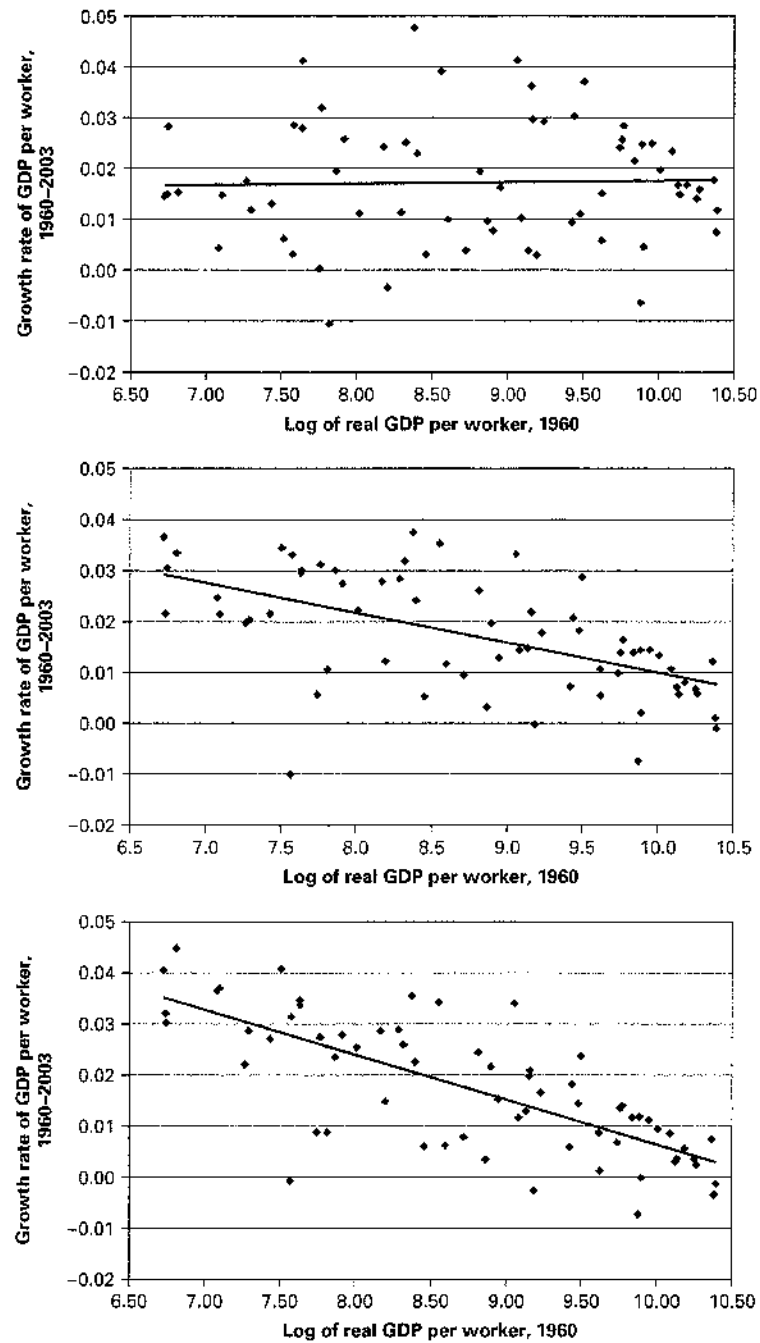
In the two lowest panels the data points do seem to be clustered around the negatively sloped straight lines, and mostly so in the bottom panel. This panel is evidence that countries that start off poor tend to grow faster (structural characteristics being equal). Particularly, in so far as the points in the bottom panel can be said to be 'represented' by one and the same straight line, it supports the hypothesis that the technological influence on growth should be roughly the same in all countries. This is a striking conclusion and we should emphasize that there is no consensus among economists concerning this view.

6.4 The importance of institutions

We have emphasized that both the steady state, level prediction, and the convergence, growth prediction of our model perform remarkably well empirically even assuming the same influence from technology across countries. We will now take a deeper look at the issue of whether technological influence can really be assumed to be similar in rich and poor countries. We will proceed as follows.⁷

First, we will focus on explaining differences in *levels* of income per worker, not in growth rates. After all, it is the level of income that is important for welfare through its impact on consumption. Income growth can be seen as a means of achieving a high level of income. Second, we will consider only the world's richest and poorest countries, aggregating

7. This section is much inspired by the article of Robert E. Hall and Charles I. Jones, 'Why do some Countries Produce so Much More Output per Worker than Others?', *Quarterly Journal of Economics*, 114, 1999, pp. 83–116. However, our method of growth accounting based on the Mankiw–Romer–Weil model differs from the method used by Hall and Jones so our conclusions are also somewhat different. Compared to us, Hall and Jones find that a larger part of the income gap between the poorest and the richest nations must be ascribed to 'technological differences'.

FIGURE 6.7 Convergence, unconditional, conditional on s_x and n , and conditional on s_x , s_y , and n , 65 countries

Source: Penn World Table 6.2. Estimates of the investment rates in human capital are from Bernanke and Gurkaynak (2001).

TABLE 6.3 The gap between the world's richest and poorest countries

Group	y_{03}	s_K	s_H	n	A
USA	1.000	0.190	0.115	0.016	
Rich 17	0.825	0.248	0.099	0.013	
Poor 17	0.033	0.094	0.023	0.022	
Rich 17 compared to Poor 17					
Factor to be explained:	24.9				
Factor explained by:	—	2.6	4.2	1.2	1.8
USA compared to Poor 17					
Factor to be explained:	30.2				
Factor explained by:	—	2.0	4.9	1.1	2.6

Note: The countries in the group Rich 17 are USA, Ireland, Norway, Belgium, Austria, France, Netherlands, Australia, Singapore, Switzerland, Italy, United Kingdom, Canada, Denmark, Hong Kong, Sweden, Finland. The countries in the group Poor 17 are Cote d'Ivoire, Congo Rep. of, Senegal, Nigeria, Nepal, Benin, Ghana, Mali, Rwanda, Kenya, Burkina Faso, Zambia, Tanzania, Malawi, Madagascar, Ethiopia, Burundi.

Source: Computations based on Table A.

each of these groups into one observation, and eliminating countries in between. The focus will thus be entirely on understanding the difference between the world's very richest and very poorest countries. Third, this time we will not *assume* that these two groups have the same level of input productivity. Rather we will try to identify how much technological difference one must assume in order to explain the full income gap on the basis of our Solow model with human capital.

Consider Table 6.3 which has been constructed from data in Table A in the Appendix to Book One. The first row reports basic data measured as hitherto for the USA. Since GDP per worker in 2003 is measured relative to the USA, there is a '1' in the y_{03} column. In the following we will sometimes identify the rich world with the USA. Should one find this too narrow, the table's second row reports the same types of data as simple averages across the 17 richest countries in the world for which the required data were available, including the USA. This amounts to defining the rich world as countries with a GDP per worker of at least 70 per cent of that of the USA. The 16 remaining countries are listed in the note for the table. The third row reports average data for the 17 poorest countries in the world for which data were available, still excluding countries with grade D for data quality in the PWT. These countries all have GDPs per worker of 7 per cent or less of the US level and they are almost all African, see again the table's note.

In the two rows under the heading 'Rich 17 compared to Poor 17', the y_{03} column reports the ratio between average GDP per worker in the 17 richest and the 17 poorest countries. So, GDP per worker is almost a factor 25 larger in the richest part of the world as compared to the poorest. If one associates the rich world with the USA the ratio is 30, as shown under the heading 'USA compared to Poor 17'. These are the factors to be explained. Their values of 25 to 30 mean that the amount of output that it takes the workers in the poor countries one year to produce are produced in less than a fortnight in the rich.

Consider the steady state prediction of our model, given by Eq. (29) above, and assume the plausible parameter values $\alpha = \varphi = \frac{1}{3}$ and $g + \delta = 0.075$, implying that both of the exponents in Eq. (29) equal one. Writing the equation for the rich and for the poor, dividing the first by the second, and omitting time subscripts on y and A gives:

$$\frac{y^{\text{rich}}}{y^{\text{poor}}} = \frac{A^{\text{rich}}}{A^{\text{poor}}} \cdot \frac{s_K^{\text{rich}}}{s_K^{\text{poor}}} \cdot \frac{s_H^{\text{rich}}}{s_H^{\text{poor}}} \cdot \left(\frac{n^{\text{poor}} + 0.075}{n^{\text{rich}} + 0.075} \right)^2 \quad (42)$$

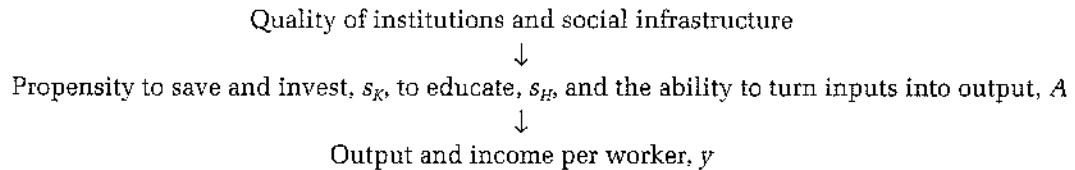
Return to Table 6.3, the section headed ‘Rich 17 compared to Poor 17’. The row labelled ‘Factor explained by’ reports under s_K the value of the ratio $s_K^{\text{rich}}/s_K^{\text{poor}}$ between the 17 richest and the 17 poorest countries. This ratio is 2.6, so differences in investment rates in physical capital can, according to our model, account for a factor 2.6 out of the factor 25 of income difference. Likewise, under s_H one finds the ratio $s_H^{\text{rich}}/s_H^{\text{poor}}$, so a factor 4.2 should be explainable by differences in investment in education. In the column labelled n , the ratio $([n^{\text{poor}} + 0.075]/[n^{\text{rich}} + 0.075])^2$ is shown, so a factor 1.2 can be attributed to differences in population (labour force) growth rates. Finally, the column labelled A shows what the ratio $A^{\text{rich}}/A^{\text{poor}}$ needs to be in order for Eq. (42) to explain the full income difference. According to our model and data, differences in ‘technology’ should account for a factor 1.8. The corresponding figures if one compares the USA to the 17 poorest countries are found in the bottom row of Table 6.3. Here, taking our model seriously, one must attribute a factor 2.6 to differences in productivity to account for the full income gap.

Roughly, income seems to differ between the richest and the poorest in the world by a factor 25–30. According to our model with reasonable parameter values, a factor 2–3 seems to be attributable to differences in (investment rates in) physical capital input, a factor 4–5 to differences in (investment rates in) human capital, and a factor 1.1–1.2 to differences in population growth rates, which leaves, finally, a factor 2–3 for differences in the productivity of the inputs of physical and human capital per worker. Our conclusion is that differences in technology are certainly of importance for income-level differences between rich and poor, but still, according to the exercise we have performed, differences in factor productivity only account for income differences to the same extent as differences in physical capital do and less than differences in human capital do. This is probably less than commonly believed.

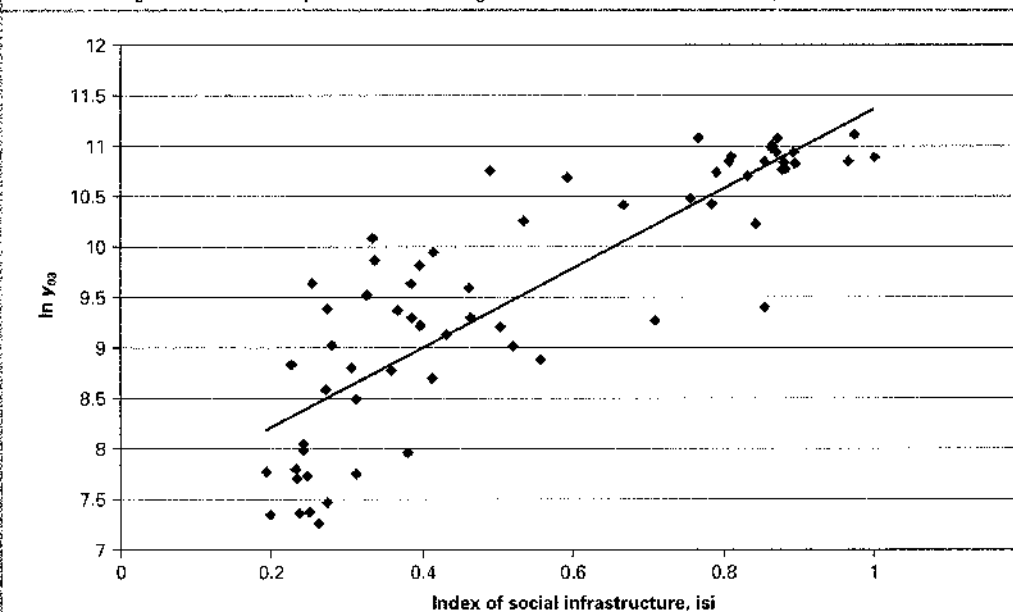
If investment rates in physical and human capital and total factor productivity are the main determinants of income levels, deeper questions arise: ‘why do some countries invest more than others in physical and human capital? And why are some countries so much more productive than others?’ These questions are quoted from the article of Hall and Jones (1999) who argue that a one-step-deeper explanation could be found, at least partly, in differences in ‘social infrastructure’ or ‘institutions’.

By social infrastructure they mean ‘the institutions and government policies that determine the economic environment within which individuals accumulate skills, and firms accumulate capital and produce output’.

To be more specific, the presence and quality of a *financial system* with safe banks and assets in which to place one’s wealth is essential to the savings and investment performance of an economy. The quality of the system of *government rules and regulations*, for instance a high degree of protection of individuals and their property, and a smooth and uncorrupted public bureaucracy in connection with starting and running businesses are important for the incentive to start up a business firm. A well-functioning system of *laws and courts* that create fast and fair decisions in case of disputes between firms and individuals promotes productivity. A sound *fiscal policy* of the government with a *fair and well-designed tax system* and an *uncorrupted expenditure side* is fundamental for the provision of essential public facilities and services such as health services and infrastructure. A good *educational system* is, of course, essential for the accumulation of skills. The general idea can be illustrated as follows:



The causality indicated by the second arrow is exactly the content of the model of this chapter with the overall conclusion that the basic parameters s_K , s_H , and A are decisive for output per

FIGURE 6.8 Logarithm of real GDP per worker in 2003 against index of social infrastructure, 64 countries

Note: The estimated line is $y = 7.43 + 3.93x$, and has $R^2 = 0.71$. The standard error of the estimate of the slope is 0.32.

Source: Penn World Table 6.2. Index of social infrastructure from data material for Hall and Jones (1999).

worker. The first arrow represents the idea that the quality of institutions affects these basic parameters. Putting it all together implies a positive influence of the quality of institutions on output per worker.

Testing such a connection involves the difficult issue of how to measure the quality of institutions, which may seem a hopeless task at first sight. However, different agencies and persons do set up indices of the quality of government, the degree of corruption, etc. across countries. Hall and Jones combine two different indices.

One comes from a private firm, Political Risk Services, that creates an 'International Country Risk Guide' rating 130 countries in a number of categories including the following five: 1) 'law and order', 2) bureaucratic quality, 3) corruption, 4) risk of expropriation, and 5) government repudiation of contracts. In each category the firm creates, from knowledge of government practices in a country, an index with a value between zero and one that is higher the more effective government policies are deemed to be. From the five indices one can define a single one by taking the average.

The other indicator is an index of openness suggested by the economists Jeffrey D. Sachs and Andrew Warner.⁸ It simply measures the fraction of years during 1950 to 1994 (a number between zero and one) that a country has been open according to some criteria, among these that average tariff rates are less than 40 per cent, that non-tariff barriers cover less than 40 per cent of trade, and that the government does not monopolize major exports. The idea is that openness and free trade are institutions that improve productivity by allowing a country to take part in international specialization and by exposing the domestic production sector to competition thereby enhancing its effectiveness.

8. Jeffrey D. Sachs and Andrew Warner, 'Economic Reform and the Process of Global Integration', *Brookings Papers on Economic Activity*, 1, 1995, pp. 1–95.

As their final index of social infrastructure Hall and Jones take the simple average of the two indices described. Figure 6.8 shows a scatter plot of GDP per worker against this combined index of social infrastructure, called '*isi*'. There is a very clear positive correlation.

The positive correlation could well be evidence that the quality of basic institutions affects income per worker positively, but as warned several times, and perhaps even more important here than elsewhere, one should be aware of the possibility of reversed causality: perhaps the positive correlation simply reflects that richer countries can afford better institutions.

Statistical methods to correct for reversed causality do exist and the article by Hall and Jones seeks to do such a correction. As a result they find an even stronger causal effect of institutions on performance than suggested by Fig. 6.2. The slope of the line of best fit in Fig. 6.8 is around four, suggesting that an increase in the *isi* of ten percentage points should give rise to a 40 per cent increase in income per worker. This is already a strong effect and if the true causal effect is even stronger, certainly institutions are very important for economic performance.

This has important implications for structural economic policy. Perhaps this should, more than anything else, be concerned with creating productive institutions such as good government conduct, protection of individuals and their property, well-functioning educational and financial systems, etc.

6.5 Summary

1. The quality and skills of the labour force are of obvious importance in human production. In this chapter we amended the Solow model to include human capital, defined as the stock of skills accumulated by workers through education and training. The stock of human capital was included among the essential inputs besides physical capital and 'raw' labour.
2. In some respects human capital was treated just like physical capital: we assumed that in each period there is a certain stock, H_t , of human capital, that human capital enters into an aggregate production function with constant returns to scale, $Y_t = K_t^\alpha H_t^\beta (A_t L_t)^{1-\alpha-\beta}$, parallel to the way physical capital enters, and that human capital accumulates by a certain fraction, s_H , of GDP being added to it each year. While empirical estimates of investment rates in physical capital are directly available from national accounts, nothing similar is true for s_H . We argued that since the most important part of investment in human capital is the GDP and income forgone while people educate themselves, the fraction of people of working age in education should give a good approximation of s_H . Using this idea, empirical estimates of s_H can be obtained across countries.
3. In one important respect human capital must be treated differently from physical capital: since human capital is embodied in persons, a firm cannot change its input of human capital and of (raw) labour independently. We made the simplifying assumption that each worker is equipped with the same amount of human capital, so each worker embodies a human capital of $h_t = H_t/L_t$, and hiring one worker means increasing raw labour input by one unit and increasing the input of human capital by h_t . Hence there is no separate market for the services of human capital: these are traded together with raw labour, and the return to human capital appears as a part of the real wage which is thus really a mix of a reward to raw labour and a return to human capital. Furthermore, in the computation of the marginal product of labour that determines the real wage, one should take human capital per worker, not total human capital, as given.

4. Our considerations on how to deal with human capital led to a model characterized by the following features:
 - The aggregate production function was as stated above. From empirical evidence on labour's share and on average wages compared to the wages of unskilled workers, we inferred that plausible parameter values could be $\alpha = \varphi = \frac{1}{3}$.
 - The prices of capital and labour were given by the marginal products.
 - The stocks of physical and human capital accumulated as implied by given investment rates, s_K and s_H , respectively, and a common depreciation rate.
 - The labour force as well as the technological variable A_t , grew at exogenously given rates.
5. The model implied convergence to a well-defined steady state in which GDP per worker follows a specific growth path with constant growth rate (equal to the exogenous growth rate of A_t). The 'steady state equation' told how the growth path depends on model parameters. According to the steady state equation:
 - An increase in s_K has a larger positive impact on GDP per worker than in the general Solow model because the rise in GDP occurring as more physical capital is accumulated will cause more human capital to be accumulated, reinforcing the increase in GDP.
 - An increase in the population growth rate has a stronger negative impact on GDP per worker than was the case in the general Solow model, because population growth now means the thinning-out of both physical and human capital.
 - An increase in s_H has a positive impact on GDP per worker similar to that of an increase in s_K .

These features were promising for an empirical test since they appeared to remedy the empirical shortcomings of the general Solow model's steady state prediction that we had identified.

6. Confronting our model's steady state prediction with cross-country empirical evidence under the heroic assumptions that the countries considered were in steady state in year 2000 and all had the same technological level in that year, we found that the model did remarkably well and actually went a long way towards solving the empirical problems with the general Solow model's steady state prediction. This was evidence that countries are perhaps not as technologically different as usually thought.
7. The steady state analysis supported policy recommendations of ensuring adequate savings and investment rates and keeping population growth rates down, the latter now being even more strongly supported than earlier. The really new issue was government policies to provide for or subsidize education. Such policies are recommendable according to our model in so far as one cannot fully trust private decision making and markets to devote a sufficient amount of resources to educational purposes. We gave several reasons why a market-determined allocation would not be optimal, implying that educational policies should be at the forefront of structural policies to promote prosperity and growth (as indeed they are in many countries).
8. The general prediction of economic performance derived from the Solow model with human capital is the convergence process implied by the model. From the model one can derive a 'convergence equation' telling how a country's average annual growth rate of GDP per worker over some period should depend on parameters and on the country's initial GDP per worker. The structure of the equation is such that the growth rate equals the underlying technological growth rate adjusted upwards for how far below steady

state the country was initially, and for the speed by which the country converges to steady state. The rate of convergence depends in a simple way on model parameters and (*ceteris paribus*) it is smaller for the Solow model with human capital than for the general Solow model. The reason is that capital accumulation takes time and delays convergence, and there is more capital to accumulate when there is also human capital. This was again promising from an empirical point of view since we have found the general Solow model to overestimate the rate of convergence.

9. Estimating the model's convergence equation on cross-country data gave a remarkably good result and in particular a much improved accordance between the estimated rate of convergence and the one predicted by the model. The good result was strongly in support of the idea of conditional convergence among the countries of the world and gave an indication that countries are perhaps also more similar with respect to their technological growth rates than usually thought.
10. Although technology and input productivity may not be as different across countries as usually believed, the empirical evidence suggests that differences in total factor productivity do contribute to explaining the income differences between rich and poor countries, along with differences in physical and human capital per worker. For explaining these factors in turn, the quality of a society's institutions and social infrastructure seems to be of great importance.

6.6 Exercises

Exercise 1. One more empirical improvement obtained by adding human capital

We opened this chapter by summarizing two main empirical shortcomings of the general Solow model. One was that its steady state prediction tends to understate the influences of saving and population growth rates on GDP per person. There may be another one that also concerns the general Solow model's steady state prediction which is (setting $ng \approx 0$):

$$\ln y_i^* = \ln A_i + \frac{\alpha}{1-\alpha} [\ln s_k - \ln(n+g+\delta)].$$

The elasticity of y_i^* with respect to s_k is the same as the elasticity with respect to $n+g+\delta$ in absolute value. Show that both are $\alpha/(1-\alpha)$, and hence should be around 0.5. To check empirically whether the two elasticities are approximately equal, construct two figures, taking data from Table A and only considering countries in the sample of 65 countries indicated in Table A (the figures are reminiscent of, but not the same as, Figs 3.7 and 3.8):

One figure should plot $\ln y_{00}^i$ against $\ln s_k^i$ across countries i . Estimate (by OLS) and draw the line of best fit through the points and identify its slope. The other figure should plot $\ln y_{00}^i$ against $\ln(n^i + 0.075)$, and include the line of best fit through the points. The slope of the line should be identified. Do the two slopes seem to be of equal size (in absolute value), or does one seem to be greater than the other? In case of the latter, which of the estimated elasticities is largest? How do your findings accord with the general Solow model's steady state prediction? How do they accord with that of the Solow model with human capital?

Another way to proceed is to do an OLS estimation of the regression equation:

$$\ln y_{00}^i = K + \gamma \ln s_k^i + \nu \ln(n^i + 0.075),$$

where we have *not* restricted the coefficients of $\ln s_k^i$ and $-\ln(n^i + 0.075)$ to be equal. If you know how to do an OLS estimation with more than one explanatory variable on the right-hand side, do an estimation of the above equation and compare your estimates of γ and ψ . Are they of about equal size in absolute value (taking uncertainty on the estimates into account)? If one is larger than the other, how does that accord with each of the two models?

Exercise 2. Government in the Solow model with human capital

When we were setting up the basic Solow model in Chapter 3 we were careful to explain that the savings rate, s , in $S_t = sY_t$ could be interpreted as $s = (1 - c^p)(1 - c^g)$, where we now assume $i_t^s = 0$, and constant values for the private propensity to consume out of disposable income, c^p , and for the spending share, c^g , of government that balanced its budget each period. The model could thus be interpreted to include a government and obviously a government could be interpreted into the general Solow model considered in Chapter 5 in exactly the same way. This exercise asks you to show how government can be included in the model of this chapter.

Assume that the government always taxes all income by the constant rate τ , and in each period spends the constant fraction s_H^g of the tax revenue on education and the remaining tax revenue on government consumption (thus balancing its budget each period and not investing in physical capital). What the government spends on education accumulates as human capital with the households.

Assume that out of their disposable incomes households invest the fraction s_K^p in physical capital and the fraction s_H^p in human capital. In all remaining respects the model is as in this chapter. Show that the model presented in this chapter can be interpreted to cover the case with a government if the parameters of the chapter are: $s_K = (1 - \tau)s_K^p$ and $s_H = \tau s_H^g + (1 - \tau)s_H^p$.

Exercise 3. Balanced growth in steady state

Show that growth in the steady state of the Solow model with human capital in *all* respects accords with the concept of balanced growth. In particular, derive an expression for the steady state real rental rate, r^* , and show that this is constant. Also compare the expression to the corresponding one in the general Solow model. How is it possible that r^* does not depend on s_H ? This is difficult to explain intuitively, but we only ask you to try to identify some counteracting factors that together can imply that r^* is not sensitive to s_H .

Exercise 4. Golden rule

Show that, given the values of the model's other parameters, the sizes of the investment rates, s_K and s_H , that will maximize the altitude of the growth path of consumption per worker in the steady state of the Solow model with human capital are $s_K = \alpha$ and $s_H = \varphi$.

Exercise 5. Further simulations to investigate stability

In Section 6.3 we ran a simulation of the Solow model with human capital to check stability of $\tilde{k}_t, \tilde{h}_t$ towards the steady state $\tilde{k}^*, \tilde{h}^*$. We considered only one starting point, $\tilde{k}_0 = 16, \tilde{h}_0 = 2$, which resulted in convergence to steady state, but in an interesting way that included some non-monotonicity ($\tilde{k}_t$ first decreases, then increases). Run some simulations yourself where (first) you consider the same parameter values as in Section 6.3, but you choose different initial points. Make diagrams like Fig. 6.6, including the two curves corresponding to constant state variables. Be sure that you have initial points in each of the figure's four regions (preferably several in each region which are spread out in a 'representative' way) and that you simulate over a sufficient number of periods to be able to judge whether there

seems to be convergence to steady state. Choose some changed, but still realistic parameter values (obeying all stated parameter restrictions), and do the same again. Do your simulations seem to support convergence to steady state generally?

Exercise 6. The effects of various parameter changes

Let a *base scenario* be defined by the economy being, in all periods from 0 to 200, say, in steady state of the Solow model with human capital at parameter values $\alpha = \varphi = \frac{1}{3}$, $s_K = 0.12$, $s_H = 0.2$, $\delta = 0.055$, $n = 0$, $g = 0.02$, and with an initial value of the technological variable $A_0 = 1$.

We will consider various *alternative scenarios*, each one defined by the economy evolving according to the base scenario from period zero through period nine, while (working first time) in period ten a permanent shift in one (or several) parameters occurs, while other unmentioned parameters stay unchanged.

For each alternative scenario the qualitative effects of the parameter shift should be described and illustrated by showing how it affects the phase diagram and by explaining intuitively the transition towards the new steady state. The quantitative effects should be computed and illustrated by conducting the appropriate simulations and creating the following figures:

- A Diagram 1 showing the evolution of $\ln y_t$ against periods $t = 0, 1, \dots, 200$ according to 1) the base scenario (the old steady state), 2) the new steady state (as if the economy had, in all periods, been in steady state at the new parameter values and with $A_0 = 1$) and 3) the alternative scenario.
- A Diagram 2 showing $\ln c_t$ the way Diagram 1 illustrates $\ln y_t$.
- A Diagram 3 showing the growth rate $g'_t = (y_t - y_{t-1})/y_{t-1}$ over periods $t = 1, \dots, 200$, otherwise as Diagram 1.
- A Diagram 4 showing K_t/Y_t and H_t/Y_t , otherwise as Diagram 1.

Alternative 1 is defined by the investment rate in physical capital increasing to $s'_K = 0.24$. In particular, what is the half-life for $\bar{y}_t$ (the number of periods it takes for $\bar{y}_t$ to go half the way from the old to the new steady state)? Compare to the half-life you found in Exercise 4 of Chapter 5, and explain the difference.

Alternative 2 is defined by the investment rate in human capital increasing to $s'_H = 0.25$.

From your two analyses comment on the development perspectives from increasing the investment rates, in particular with respect to the duration of the consumption sacrifice.

Alternative 3 is defined by the investment rate in physical capital increasing to $s'_K = 0.24$, and the investment rate in human capital decreasing to $s'_H = 0.15$.

Does the economy always seem to react to parameter changes with a 'growth jump'?

Exercise 7. Inferring α and φ from the estimation of the convergence equation

First find a way to estimate values for α and φ from the information gathered in (39), (40) and (41). Then show that the implied values are $\alpha = 0.39$ and $\varphi = 0.30$. Comment.

Exercise 8. Illustrating further the improvement of the convergence equation obtained by adding human capital

In this chapter we created Figs 6.4 and 6.5 to illustrate the improvement in the steady state equation of the model with human capital over that of the model without human capital. A line of best fit should have a slope around one for the model to be correct. The line in the model with human capital had such a slope, but the model without human capital did not. This exercise asks you to make a similar illustration with respect to the convergence equations. You should consider the same 65 countries considered in Figs 6.4 and 6.5. Data can be taken from Table A at the back of Book One.

1. First consider the convergence equation of the general Solow model:

$$g_{T,0}^y \equiv V - \frac{1-(1-\lambda)^T}{T} \ln y_0 + \frac{1-(1-\lambda)^T}{T} \cdot \frac{\alpha}{1-\alpha} [\ln s_K - \ln(n+g+\delta)],$$

where we have denoted the 'technological influence' by V and set $ng = 0$. Assuming as usual $\alpha = \frac{1}{3}$ and $n+g+\delta = 0.075$ (and remembering $\lambda = (1-\alpha)(n+g+\delta)$), what are the 'theoretical' sizes of the two coefficients, $\beta_1 = (1-(1-\lambda)^T)/T$ and $\beta_2 = \beta_1\alpha/(1-\alpha)$, in this equation if $T = 43$? In a figure, plot $g_{03,60}^i$ against $-\beta_1 \ln y_{60}^i + \beta_2 [\ln s_K^i - \ln(n^i + 0.075)]$ with β_1 and β_2 so computed and check if the line of best fit seems to have a slope of one.

2. Then consider the convergence equation of the Solow model with human capital:

$$g_{T,0}^y \equiv V - \frac{1-(1-\lambda)^T}{T} \ln y_0 + \frac{1-(1-\lambda)^T}{T} \frac{\alpha}{1-\alpha-\varphi} [\ln s_K - \ln(n+g+\delta)] \\ + \frac{1-(1-\lambda)^T}{T} \frac{\varphi}{1-\alpha-\lambda} [\ln s_H - \ln(n+g+\delta)].$$

Assuming $\alpha = \varphi = \frac{1}{3}$ and $n+g+\delta = 0.075$ (and remembering $\lambda = (1-\alpha-\varphi)(n+g+\delta)$), compute the 'theoretical' sizes of the three coefficients β_1 , β_2 and β_3 in the equation given $T = 43$. In a figure, plot $g_{03,60}^i$ against $-\beta_1 \ln y_{60}^i + \beta_2 [\ln s_K^i - \ln(n^i + 0.075)] + \beta_3 [\ln s_H^i - \ln(n^i + 0.075)]$ and check if the line of best fit seems to have a slope of one.

Exercise 9. An alternative Solow model with human capital

You may have been wondering why we have built human capital into the model of this chapter in a way that is different from how it was brought into the picture in Section 5.5 on growth accounting. In fact, there is an alternative growth model with human capital that incorporates human capital in the way it was done in Chapter 5.

This alternative model with human capital consists of the same equations as the general Solow model, that is, Eq. (7)–(13) of Chapter 5, with the following exceptions: the production function (7) is changed to:

$$Y_t = K_t^\alpha (A_t h L_t)^{1-\alpha},$$

where:

$$h = \exp(\psi u),$$

and ψ is a given productivity parameter in the production of human capital and u is the average number of years that people in the labour force have been to school (the same in all periods). The 'rental rate equal to marginal product' conditions (8) and (9) should be modified appropriately, but are not used in the following.

Both ψ and u are positive and constant parameters and to be thought of as rates like α and s , respectively. Realistically ψ should be around 0.1 (see again Section 5.5). Note that in this model human capital per worker, h , is simply a constant. The difference between this alternative model and the one considered in this chapter is that in the alternative model human capital per worker is not being built up gradually through an accumulation process where additions come from investing a given fraction of GDP each year, but rather is simply a given constant throughout. Nevertheless, how this constant depends on education as measured by u may be important.

The full model is conveniently analysed in terms of the auxiliary variables $\tilde{y}_t = y_t/(hA_t)$ and $\tilde{k} \equiv k_t/(hA_t)$, where y_t and k_t are output per worker, Y_t/L_t , and capital per worker, K_t/L_t , respectively.

1. Show that $\bar{y}_t = \bar{k}_t^\alpha$ and that the law of motion for $\bar{k}_t$ is:

$$\bar{k}_{t+1} = \frac{1}{(1+n)(1+g)}(s\bar{k}_t^\alpha + (1-\delta)\bar{k}_t).$$

2. Show that $\bar{k}_t$ and $\bar{y}_t$ converge to constant steady state values and that in steady state output per worker evolves as:

$$y_t^* = A_t \left(\frac{s}{n+g+\delta+ng} \right)^{\alpha/(1-\alpha)} \exp(\psi u).$$

3. The 'steady state equation' derived from the above is:

$$\ln y_t^* = \ln A_t + \frac{\alpha}{1-\alpha} [\ln s - \ln(n+g+\delta+ng)] + \psi u.$$

Test this empirically as follows. Consider our sample of 65 countries for which also data on u^i are available from Table A. Consider first the model without human capital ($\psi = 0$). Assuming $\alpha = \frac{1}{3}$, plot $\ln y_{03}^i$ against $0.5[\ln s^i - \ln(n^i + 0.075)]$ across countries i . This is almost what we did in Fig. 6.4, but not for exactly the same countries. Estimate the line of best fit and judge if it seems to have a slope of one. Next, plot $\ln y_{03}^i$ against $0.5[\ln s^i - \ln(n^i + 0.075)] + 0.1u^i$. Estimate again the line of best fit and see if it has come closer to having a slope of one. Is the improvement of the steady state prediction comparable to the one obtained in the model of this chapter?

4. By linearizing the above law of motion around steady state, etc., show that the 'convergence equation' of this model is:

$$\frac{\ln y_T - \ln y_0}{T} \cong g + \frac{1-(1-\lambda)^T}{T} \times \left(\ln A_0 + \frac{\alpha}{1-\alpha} [\ln s - \ln(n+g+\delta+ng)] - (\ln y_0 - \psi u) \right),$$

where $\lambda \equiv (1-\alpha)(n+g+\delta)$. A regression equation suggested by this convergence equation is:

$$g_{03,80}^i = \beta_0 - \beta_1(\ln y_{60}^i - 0.1u^i) + \beta_2[\ln s^i - \ln(n^i + 0.075)].$$

Estimate this by OLS, taking data from Table A. Are the estimated parameters significant and do they have the right signs? What are the values for α and λ that can be derived from the estimated parameters? By introducing human capital, does this model bring a better accordance between the model's rate of convergence (an evaluation of λ for realistic parameter values) and the estimated rates as compared to what was found in the general Solow model? Is the improvement comparable to the one obtained in the chapter's model with human capital? Why not?

5. Discuss the advantages and disadvantages of the two models with human capital. For instance, which data do you think are most reliable, those for u^i used in the model of this exercise or those for s_H^i used in this chapter? Which of the models do you think has the nicest properties? Is there a trade-off with respect to which model to prefer?

Exercise 10. An empirical shortcoming of the Solow model with human capital?

Exercise 1 above pointed to an(other) empirical shortcoming of the Solow model without human capital. The model's prediction that the investment rate in physical capital and the population growth rates should affect GDP per worker equally strongly (but in opposite directions) does not seem to hold true empirically. Rather, there seems to be a stronger influence from the population growth rate than from the investment rate, a feature indeed found in the Solow model with human capital. Let's now take this latter model through a

similar direct challenge. The steady state prediction of the Solow model with human capital is (setting $ng \cong 0$):

$$\ln y_t^* \cong \ln A_t + \frac{\alpha}{1-\alpha-\varphi} [\ln s_K - \ln(n+g+\delta)] + \frac{\varphi}{1-\alpha-\varphi} [\ln s_H - \ln(n+g+\delta)].$$

Find the elasticities of y_t^* with respect to s_K , s_H and $n+g+\delta$, respectively. Show that if $\alpha = \varphi = \frac{1}{3}$ (as we more or less believe) then the elasticities with respect to s_K and s_H should be of equal size and both should be one, while the elasticity with respect to $n+g+\delta$ should be twice as large in absolute value.

Taking data from Table A and considering the sample of 65 countries, do an OLS estimation of the following regression equation (requires that you can do an OLS estimation with more than one explanatory variable on the right-hand side):

$$\ln y_{90}^i = K + \gamma \ln s_K^i + \varepsilon \ln s_H^i + \nu \ln(n^i + 0.075),$$

where we have *not* imposed any model-based restrictions on the coefficients to $\ln s_K^i$, $\ln s_H^i$, $\ln(n^i + 0.075)$. How do your estimates of γ , ε and ν accord with what the elasticities should be? Take the uncertainty of your estimates into account when you answer.

Exercise 11. Institutions and the intermediate explanatory variables

According to the reasoning of Section 6.4, the quality of institutions should be important for the ‘intermediate’ explanatory variables s_K , s_H and A , which in turn should be important for y . We tested this in Fig. 6.8 by plotting y against isi , finding a clear positive correlation. For the theory to be correct there should, however, be a positive influence of isi on each of the variables s_K , s_H and A , which was not tested. This exercise seeks to take a step in the direction of such a test.

Consider Table A and the 64 countries that are in our sample of 65 countries and for which the isi is available (we don’t have the isi for Nepal). Let the USA be ‘country of comparison’.

1. For each of the remaining 63 countries decompose the income per worker relative to the USA, y^i/y^{US} , into components the way it was done for rich vs. poor in Section 6.4, thereby among other things creating relative technology levels, A^i/A^{US} , for the 63 countries other than the USA.
2. Plot y^i/y^{US} against A^i/A^{US} across the 63 countries and comment with respect to whether technological differences seem to be important for understanding income differences.
3. In three figures, plot each of s_K^i , s_H^i and A^i/A^{US} against isi across the 63 countries. Comment with respect to the issue of this exercise.